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A Introduction

Goal of the work, maybe a simple example, some mention to preliminary information and a soft mention to what's the problem and the philosophy.

B Minkowski space and space-like surfaces

{sc:splsrf}

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ endowed with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish the use of this metric from the standard euclidean metric of $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates either as (x^1, x^2, x^3) or (x, y, t), choosing the notation according to convenience and will identify the slice $\{(x, y, 0)\}$ of $\mathbb{M}^{2,1}$ with $\mathbb{R}^2$ and with $\mathbb{C}$.

The space $\mathbb{M}^{2,1}$ is an example of a more general object called a pseudo-Euclidean vector space, a term that refers to a finite-dimensional vector space V endowed with a nondegenerate (possibly not positive-definite) symmetric bilinear form g. This kind of space serves as the "infinitesimal model" for pseudo-Riemannian manifolds, smooth manifolds endowed with a smooth nondegenerate 0-2 tensor. If the signature of this tensor is exactly $(n-1, 1)$, the manifold is called a Lorentzian manifold.
The following definition, borrowed from physics, will be useful later.

Definition. Given (M, g) a Lorentzian manifold and $p \in M$, we say that a vector $v \in T_pM$ is

- space-like if $\langle v, v\rangle > 0$,
- light-like if $\langle v, v\rangle = 0$ and $v \neq 0$,
- time-like if $\langle v, v\rangle < 0$.

We also call a subspace $P \subseteq T_pM$ space-like if every nonzero vector in P is space-like.

We will always denote with S a differentiable manifold of dimension 2 (a surface) immersed in $\mathbb{M}^{2,1}$, $\iota : S \hookrightarrow \mathbb{M}^{2,1}$. We assume that the ambient pseudometric, $g_{\mathbb{M}^{2,1}}$ when restricted to S, yields a nondegenerate positive definite bilinear form. We call this form the first fundamental form or pull-back pseudometric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and write in coordinates as $\mathbf{I} = g_{ij}dx^i dx^j$ (for simplicity we sometimes forget the immersion ι).

Space-like immersions.
The objects we will focus on are not arbitrary immersed surfaces, but the so-called space-like surfaces of $\mathbb{M}^{2,1}$. This term refers to immersed surfaces in $\mathbb{M}^{2,1}$ whose induced metric is positive-definite, and hence Riemannian. Equivalently, these are the immersed surfaces S such that, for every $p \in S$, the tangent plane T_pS is space-like.

A nice property of this kind of surface is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$, f being smooth and Lipschitz with constant less than 1.
We can show this with a simple argument:
First, we assumed that the restriction of the Lorentzian metric to each tangent space T_pS is positive-definite so that it is a space-like plane. This can be

rewritten as

$$T_pS \subset \mathscr{C} := \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle \geq 0\} = \{t^2 \leq x^2 + y^2\} \ .$$

The set $\mathscr{C}$ is the outside-region of the double cone in $\mathbb{M}^{2,1}$ whose axis is $\{x^1 = 0,\ x^2 = 0\}$. Clearly $\mathscr{C}$ does not contain any vertical vector and clearly the same holds for T_pM. Therefore we can apply the implicit function theorem and express S locally near p as the graph of a function $t = f(x, y)$, where $f : U \subset \mathbb{R}^2 \to \mathbb{R}$. Where this local representation is valid, we can parametrize the surface as $(x, y, f(x, y))$, and obtain the local frame

$$V_x = (1, 0, \partial_x f(x, y)), \qquad V_y = (0, 1, \partial_y f(x, y)),$$

for which the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x\rangle & \langle V_x, V_y\rangle \\ \langle V_y, V_x\rangle & \langle V_y, V_y\rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

Since the parametrization is regular, the condition that g be positive-definite is equivalent to

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\nabla f|^2,$$

and therefore $|\nabla f|^2 < 1$.

Entire surfaces.
Given a surface S immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$. We call the surface S is entire when the map Π is a diffeomorphism. If the surface is globally a graph of a function φ, this condition is exactly stating that the function φ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for complete space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global as we prove in the proposition below.

{prop:toppropr spimm}

Proposition 1. *Let Ω be a domain in $\mathbb{R}^2$ (that is, an open connected set), let $S \overset{\iota}{\hookrightarrow} \mathbb{M}^{2,1}$ be a space-like immersion and $\psi : \Omega \to S$ a parametrization of S. Suppose that S is complete with respect to the induced Riemannian metric. Then S is an entire graph and Ω is simply connected.*

Before laying out the proof let us recall a definition and a proposition from the theory of coverings. Let M and N two differentiable manifolds and $f : M \to N$ a smooth map.

Definition B.1. We say that the map f satisfies the path lifting property if for every $p \in N$ and for every path $\alpha : I \to M$ such that $\alpha(0) = f(p)$ there exists a path $\tilde{\alpha} : I \to N$ such that $\tilde{\alpha}(0) = p$ and $f \circ \tilde{\alpha} = \alpha$. We can assume without loss of generality that the paths involved are differentiable.

Proposition 2. *If f is a local homeomorphism and satisfies the path lifting property, then it is a covering map.*

Proof of Proposition 1. The idea of the proof is to show that $\Pi \circ \iota$ is a global diffeomrphism. Clearly this implies that Π restricted to S is a diffeomeorphism on $\mathbb{R}^2$ and so that S is an entire graph. From that we obtain also that Ω is simply connected as $\Pi \circ \iota \circ \psi$ is a diffeomorphism with $\mathbb{R}^2$.

Let us observe that the target of $\Pi \circ \iota$ is just $\mathbb{R}^2$, which is simply connected, and the domain is Ω, which is connected since it is a domain. We already proved, by showing that S is locally a graph, that the map $\Pi \circ \iota$ is a local homeomorphism.
Having met these conditions, if we show that $\Pi \circ \iota$ satisfies the path lifting property, we conclude that it is a smooth covering map whose domain is simply connected, hence a diffeomorphism.

Let us fix a point $z \in \Omega$ and a path $\gamma : [0,1] \to \mathbb{R}^2$ of bounded length, $\ell(\gamma) < \infty$, such that $\gamma(0) = (\Pi \circ \iota)(z)$. Our goal is to show that γ admits a lift to a path γ^* on Ω starting at z.
To this end, let us consider the set of times up to which such a lift exists,

$$\mathcal{J} := \{\tau \in [0,1] \mid \exists \gamma^* : [0,t] \to \Omega \text{ such that } (\Pi \circ \iota) \circ \gamma^* = \gamma \text{ and } \gamma^*(0) = z\},$$

and let us define $\bar{\tau} := \sup \mathcal{J}$.

Let us observe that $0 \in \mathcal{J}$ and that $\mathcal{J}$ is open. This is true because, being $\Pi \circ \iota$ a local homeomorphism, if a certain τ is in $\mathcal{J} \cap (0,1)$ then the whole interval $(\tau - \varepsilon, \tau + \varepsilon)$ for $\varepsilon > 0$ sufficiently small, belongs to it.

If we prove that $\bar{\tau} = 1$, then γ admits a lift on the whole interval $[0,1]$ and we have finished. Then let us suppose by contradiction that $\bar{\tau} < 1$ and define $\gamma^* : [0, \bar{\tau}) \to \Omega$ the lift of γ up to $\bar{\tau}$. This curve exists by definition of $\bar{\tau}$.

Finally, we need to prove that the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(t)$ exists. If this is true we can then extend γ^* up to $\bar{\tau}$, which means that $\bar{\tau} \in \mathcal{J}$. Since $\mathcal{J}$ is open, there exists an element $\bar{\tau} + \varepsilon \in \mathcal{J}$, $\varepsilon > 0$. This contradicts the definition of $\bar{\tau}$ and proves $\bar{\tau} = 1$ since the alternative is absurd.

To prove that $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists we just need a couple of observations:
First we note that the map Π dilates distances, as

$$\begin{aligned} \langle v, v \rangle_{\mathbb{M}^{2,1}} &= (v^1)^2 + (v^2)^2 - (v^3)^2 \\ &\leq (v^1)^2 + (v^2)^2 \\ &= \langle \Pi v, \Pi v \rangle_{\mathbb{R}^2} . \end{aligned}$$

From this it is clear that $\ell(\gamma^*) \leq \ell(\gamma)$.
Now, if we fix any sequence $\tau_k \to \bar{\tau}$ and observe that as γ is differentiable, it is derivative is bounded, so $\ell(\gamma|_{[\tau_k, \tau_m]}) \leq C|\tau_k - \tau_m|$. By observing that

$$\begin{aligned} d(\gamma^*(\tau_k), \gamma^*(\tau_m)) &\leq \ell(\gamma^*|_{[\tau_k, \tau_m]}) \\ &\leq \ell(\gamma|_{[\tau_k, \tau_m]}) \\ &\leq C|\tau_k - \tau_m| \end{aligned}$$

we deduce that also $(\gamma^*(\tau_k))_k$ is a Cauchy sequence, hence converging since S is complete and Ω is to be intended with the induced metric.
By the arbitrariness of the sequence $(\tau_k)_k$ we deduce the limit $\lim_{\tau\to\bar\tau}\gamma^*(\tau)$ exists. $\square$

In this section we have essentially proved that a complete space-like surface immersed in $\mathbb{M}^{2,1}$ is actually the graph of a smooth function f satisfying $|\nabla f|^2 < 1$. A partial converse also holds: the graph of a smooth function whose gradient has norm strictly smaller than 1 is a space-like surface, although it may lack completeness.

Normal vectors to a space-like surface in $\mathbb{M}^{2,1}$
Since the ambient metric g is nondegenerate and each tangent plane T_pS is space-like, the restriction of g to T_pS is positive-definite, hence nondegenerate. It follows that, for every $p \in S$,

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS \ .$$

Since S has codimension 1 in $\mathbb{M}^{2,1}$, the orthogonal complement $(T_pS)^\perp$ is one-dimensional. Therefore it is generated by a vector $N = N_p$ normal to S.

We can easily show that N must be a time-like vector, since T_pS is space-like.
Let us fix $e_1, e_2 \in T_pM$ two independent vectors. Since N belongs to $(T_pM)^\perp$ we have $\langle e_1, N\rangle = \langle e_2, N\rangle = 0$. This implies that when we look at the representative matrix of g_p written in the basis (N, e_1, e_2), we have

$$[g_p] = \begin{pmatrix} \langle N,N\rangle & 0 & 0 \\ 0 & \langle e_1,e_1\rangle & \langle e_1,e_2\rangle \\ 0 & \langle e_2,e_1\rangle & \langle e_2,e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle N,N\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form g restricted to the subspace T_pM. We conclude that $\langle N, N\rangle$ is negative just by observing that $\det[g_p] < 0$ as g is a Lorentzian metric, while $\det[(g|_S)_p] > 0$ since g becomes a metric when restricted to S.

Since $(T_pS)^\perp$ is 1-dimensional, all of its vectors are proportional to each other. Moreover, as we have just seen, they are all timelike, so if w is a nonzero vector in $(T_pS)^\perp$, then

$$\langle w, w\rangle < 0.$$

We can thus rescale w by a factor $k \in \mathbb{R}$ such that $\langle kw, kw\rangle = -1$. This equation yields two possible choices of k, one for each connected component of the space $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1\}$. We can consistently choose the connected component with positive time

$$\mathbb{H} = \mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1,\ v^3 > 0\} \ ,$$

effectively selecting a unique normal vector for each $p \in S$. The choice of the component within $\{v^3 > 0\}$ or $\{v^3 < 0\}$ is sometimes referred to a "time-orientation" for the space $\mathbb{M}^{2,1}$, and can be performed independently from the surface S. This way we obtain a globally defined map

$$\tilde{u} : S \to \mathbb{H} \ . \tag{1}$$ {eq:gausshmap}

{sc:Hmodel}

$\mathbb{H}$ is a model for the Hyperbolic plane.
The space $\mathbb{H}$ that we just defined is clearly a surface immersed in $\mathbb{M}^{2,1}$, but there is more. First of all, we observe that, by manipulating the definition of $\mathbb{H}$, we can write it as the graph of the function $\varphi(x,y) = \sqrt{x^2+y^2+1}$. The gradient of such function is $\nabla\varphi = \frac{1}{\sqrt{x^2+y^2+1}}(x,y)$, from which it is clear that $|\nabla\varphi| < 1$. By applying the result we proved before, we obtain that the bilinear form that $\mathbb{H}$ inherited from the immersion in $\mathbb{M}^{2,1}$ is a Riemannian metric. In fact this space is one of the many models of the hyperbolic plane.
Another model for the same Riemannian manifold is the Poincaré disk. As a differentiable manifold, the Poincaré disk is simply the unit ball in $\mathbb{R}^2$, $\mathbb{D} = \{\xi \in \mathbb{R}^2 \mid |\xi| \leq 1\}$, while we can construct the metric by exhibiting a diffeomorphism between $\mathbb{D}$ and $\mathbb{H}$ pulling back the metric.
A possible diffeomorphism is the map p, whose inverse is the map q, as can be verified by direct computation:

$$\begin{aligned} p\colon \mathbb{H} &\longrightarrow \mathbb{D} & q\colon \mathbb{D} &\longrightarrow \mathbb{H} \\ (x,y,t) &\longmapsto \frac{1}{1+t}(x,y) & \xi &\longmapsto \frac{1}{1-|\xi|^2}(2\xi^1, 2\xi^2, 1+|\xi|^2). \end{aligned}$$

It is clear that both functions are smooth, since $|\xi| < 1$ and $t > 1$.

We would like to find the expression of $q^*(g_{\mathbb{H}})$.To simplify the calculations we might observe that the metric of $\mathbb{H}$ inherited by $\mathbb{M}^{2,1}$ is invariant under a rotation on the t axis, and that such a rotation corresponds under p to a rotation of the disk. This means that the metric $q^*(g_H)$ will be invariant under rotation. It is then sufficient to fix a certain $\xi \in \mathbb{D}$ of the form $\xi = (a,0)$ and calculate $q^*(g_{\mathbb{H}})_\xi$; this will specify the whole metric tensor. The matrix associated to dq_ξ in the canonical bases $\mathscr{B}_1 = (e_1, e_2)$ of $\mathbb{R}^2$ and of $\mathscr{B}_2 = (E_1, E_2, E_3)$ of $\mathbb{M}^{2,1}$ is

$$[dq_\xi]_{\mathscr{B}_1\mathscr{B}_2} = \frac{2}{(1-|\xi|^2)^2}\begin{pmatrix} 1+(\xi^1)^2-(\xi^2)^2 & 2\xi^1\xi^2 \\ 2\xi^1\xi^2 & 1-(\xi^1)^2+(\xi^2)^2 \\ 2\xi^1 & 2\xi^2 \end{pmatrix}.$$

Which, evaluated at $\xi = (a,0)$, yields

$$[dq_{(a,0)}]_{\mathscr{B}_1\mathscr{B}_2} = \frac{2}{1-a^2}\begin{pmatrix} \frac{1+a^2}{1-a^2} & 0 \\ 0 & 1 \\ \frac{2a}{1-a^2} & 0 \end{pmatrix} =: \frac{2}{1-a^2}\left(v_1|v_2\right).$$

We observe that, with respect to the metric of $\mathbb{M}^{2,1}$, it holds

$$\langle v_1, v_1\rangle = 1, \qquad \langle v_2, v_2\rangle = 1, \qquad \langle v_1, v_2\rangle = 0.$$

Therefore $\mathscr{B}_3 = (v_1, v_2)$ is an orthonormal basis of $T_{q(0,a)}\mathbb{H}$ with respect to $g_{\mathbb{H}}$, hence $[g_{\mathbb{H}}]_{\mathscr{B}_3} = I$, while $dq_{(a,0)}$ represented by

$$[dq_{(a,0)}]_{\mathscr{B}_3\mathscr{B}_1} = \frac{2}{1-a^2}I.$$

So we have

$$\begin{aligned}[q^*(g_{\mathbb{H}})_{(0,a)}]_{\mathscr{B}_1} &= [dq_{(0,a)}]^T_{\mathscr{B}_3\mathscr{B}_1}[g_{\mathbb{H}}]_{\mathscr{B}_3}[dq_{(0,a)}]_{\mathscr{B}_3\mathscr{B}_1} \\ &= [dq_{(0,a)}]^T_{\mathscr{B}_3\mathscr{B}_1} I [dq_{(0,a)}]_{\mathscr{B}_3\mathscr{B}_1} \\ &= \left(\frac{2}{1-a^2}\right)^2 I \quad .\end{aligned}$$

In particular, $q^*(g_{\mathbb{H}})_{(0,a)}$ is a multiple of the Euclidean metric.
By the argument on rotational invariance, we conclude that for every $\xi \in \mathbb{D}$,

$$q^*(g_{\mathbb{H}})_\xi = \left(\frac{2}{1-|\xi|^2}\right)^2 g^E_\xi .$$

The disk $\mathbb{D}$ with this metric, which we denote also with ds^2_p, is called the Poincaré model of the Hyperbolic plane.

The Gauss Map in $\mathbb{M}^{2,1}$
In (1) we constructed a map $\tilde{u}$ between the surface S and $\mathbb{H}$. Having now at our disposal the isometry q between $\mathbb{H}$ and the Poincaré disk, $\mathbb{D}$, we can compose $\tilde{u}$ with q, obtaining what is called the Gauss map of the surface S.

$$u := q \circ \tilde{u} : S \to \mathbb{D}.$$

The construction we followed is analogous to the classical one, except that the target of the Gauss map is here $\mathbb{D}^2$ instead of $\mathbb{S}^2$. Moreover, unlike the Euclidean case, for a space-like surface we always have a globally defined unit normal field: once a time orientation of $\mathbb{M}^{2,1}$ is fixed, the timelike line $(T_pS)^\perp$ intersects the corresponding connected component of $\hat{\mathbb{H}}^2$ in exactly one point for every $p \in S$. Instead, for an immersed surface in Euclidean space, there is no canonical way to choose a normal vector.

C The Gauss-Codazzi equations and the curvatures of S

In this section we derive the Gauss-Codazzi equations for a space-like surface immersed in $\mathbb{M}^{2,1}$ and see that they assume a different form from the Euclidean case. These equations will play a central role in the discussion that follows, especially due to Bonnet's fundamental theorem on surfaces, stated at the end of the section.

C.1 Recall on the second fundamental form

We repeat now the construction of the second fundamental form of the immersion ι, which brings a different sign convention that will occur also to the expression of the Gaussian curvature.

As we have already noted, given a point $p \in S$, since S is a space-like surface, we can decompose the tangent space

$$T_p\mathbb{M}^{2,1} = T_pS \oplus (T_pS)^\perp \ .$$

This splitting is differentiable, in the sense that the mappings

$$(p, v) \mapsto (p, v^T) \qquad \text{and} \qquad (p, v) \mapsto (p, v^N)$$

are differentiable maps of $T\mathbb{M}^{2,1}|_S$ to $T\mathbb{M}^{2,1}|_S$.

Denote with D the Levi-Civita connection on $\mathbb{M}^{2,1}$ induced by the pseudo-metric $g_{\mathbb{M}^{2,1}}$ and with ∇ the connection induced on the surface S. As usual, if X and Y are local vector fields on S with local extensions $\bar{X}$ and $\bar{Y}$ then the induced connection on S is simply the projection of the connection of the ambient space onto the surface S

$$\nabla_X Y = \left(D_{\bar{X}}\bar{Y}\right)^T \ .$$

Which means that the difference between the two connections of X and Y will be a local vector field on $\mathbb{M}^{2,1}$ normal to S,

$$B(X, Y)_p := (D_{\bar{X}}\bar{Y} - \nabla_X Y)_p \perp T_pS. \ . \tag{2}$$ {eq:defB}

As in the Euclidean case, B is a symmetric tensor and is independent of the chosen extensions $\overline{X}$ and $\overline{Y}$. Since the normal space $(T_pS)^\perp$ is one-dimensional, once a local unit normal vector field N along S is fixed, there exists a unique symmetric bilinear form $\mathbf{II} = \mathbf{II}(X, Y)$ such that

$$B(X, Y)_p = (\mathbf{II}(X, Y)N)_p,$$

From now on we forget to distinguish a vector field X with its extension $\overline{X}$. Let N be local normal vector field on S and V, W two local tangent vector fields to S. We observe first that

$$\langle N, W\rangle \equiv 0 \ ,$$

so that by taking the derivative along V,

$$\langle D_V N, W\rangle + \langle N, D_V W\rangle = 0 \ .$$

Then

$$\begin{aligned}
\langle D_V N, W\rangle &= -\langle N, B(V, W) + \nabla_V W\rangle \\
&= -\langle N, B(V, W)\rangle \\
&= -\langle N, \mathbf{II}(V, W)N\rangle \\
&= -\langle N, N\rangle\, \mathbf{II}(V, W) \\
&= \mathbf{II}(V, W) \ .
\end{aligned} \tag{3}$$ {eq:gc7}

With this choice we have, from definition 2,

$$\boxed{D_V X = \nabla_V X + \mathbf{II}(V, X)N} \ . \tag{4}$$ {eq:gc1}

Now, having fixed $W_p \in T_pS$, and since the metric on T_pS is a nondegenerate bilinear form, we can associate to the functional $\mathbf{II}(\cdot, W_p) : T_pS \to \mathbb{R}$ a unique vector $\mathbf{II}(\cdot, W_p)^\sharp \in T_pS$ such that for every $V_p \in T_pS$,

$$\left\langle \mathbf{II}(\cdot, W_p)^\sharp, V_p \right\rangle = \mathbf{II}(V_p, W_p) \ .$$

We then define the shape operator by

$$A_p W_p := \mathbf{II}(\cdot, W_p)^\sharp.$$

Thus

$$\begin{aligned} \langle A_p W_p, V_p \rangle &= \mathbf{II}(V_p, W_p) \\ &= \mathbf{II}(W_p, V_p) \\ \text{[(3)]} \qquad &= \left\langle D_{W_p} N, V_p \right\rangle . \end{aligned}$$

Since this holds for every $V_p \in T_pS$, we obtain

$$A_p W_p = D_{W_p} N.$$

So we have $A = DN$.

C.2 The Gauss-Codazzi equations and Gaussian curvature

It is easy to see that $\mathbb{M}^{2,1}$ is actually flat with its metric. This mean that its curvature tensor is zero $\overline{R}(X,Y)Z \equiv 0$. Let us rewrite $\overline{R}$ in relation with R the curvature tensor of the surface S. By definition

$$\overline{R}(X,Y)Z = D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z \quad .$$

Let us focus first on the term $D_X D_Y Z$, applying the equation (4) we get

$$\begin{aligned} D_X D_Y Z &= D_X(\nabla_Y Z + \mathbf{II}(Z,Y)N) \\ &= \nabla_X \nabla_Y Z + \mathbf{II}\left(\nabla_Y, X\right) N + d_X(\mathbf{II}(Z,Y))N + \mathbf{II}(Z,Y) D_X N \quad . \end{aligned}$$

And analogously

$$D_Y D_X Z = \nabla_Y \nabla_X Z + \mathbf{II}\left(\nabla_X, Y\right) N + d_Y(\mathbf{II}(Z,X))N + \mathbf{II}(Z,X) D_Y N \quad .$$

Finally we have, by noting that the connection ∇ is symmetric,

$$\begin{aligned} D_{[X,Y]} Z &= \nabla_{[X,Y]} Z - \mathbf{II}([X,Y], Z)N \\ &= \nabla_{[X,Y]} Z + \mathbf{II}(Z, \nabla_X Y)N - \mathbf{II}(Z, \nabla_Y X)N \quad . \end{aligned}$$

So that, by putting everything together we obtain,

$$\begin{aligned} 0 &= \overline{R}(X,Y)Z \\ &= D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z \\ &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \\ &\qquad + d_X(\mathbf{II}(Z,Y))N - \mathbf{II}(\nabla_X Z, Y)N - \mathbf{II}(Z, \nabla_X Y)N \\ &\qquad - [d_Y(\mathbf{II}(Z,X))N - \mathbf{II}(\nabla_Y Z, X)N - \mathbf{II}(Z, \nabla_Y X)N] \\ &\qquad + \mathbf{II}(Z,Y) D_X N - \mathbf{II}(Z,X) D_Y N \quad . \end{aligned} \tag{5}$$

{eq:gc2}

Now, we recognize in the expression that $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$, and by definition, we set the derivative of the tensor,

$$(\nabla_X(\mathbf{II}))(Y,Z) = d_X(\mathbf{II}(Z,Y)) - \mathbf{II}(\nabla_X Z, Y) - \mathbf{II}(Z, \nabla_X Y) \ .$$

So, by continuing in (5),

$$\begin{aligned} 0 = R(X,Y)Z + (\nabla_X \mathbf{II})(Y,Z)N - (\nabla_Y \mathbf{II})(Z,X)N \\ + \mathbf{II}(Z,Y)D_X N - \mathbf{II}(Z,X)D_Y N \ . \end{aligned} \tag{6}$$ {eq:gc3}

Now we observe that from $\langle N, N\rangle \equiv -1$ we have, for v tangent to S,

$$2\langle D_v N, N\rangle = 0$$

so that $D_v N$ is a tangent vector on S. Now, in (6) we can separate the terms that are tangent to S and those that are normal. This way we obtain two distinct equations

$$\begin{cases} R(X,Y)Z = -\mathbf{II}(Z,Y)D_X N + \mathbf{II}(Z,X)D_Y N \\ (\nabla_X \mathbf{II})(Y,Z) - (\nabla_Y \mathbf{II})(Z,X) = 0 \end{cases} \ . \tag{7}$$ {eq:gc4}

These equations are (a form of) the Gauss-Codazzi equations.

Let us take the first equation of (6) and test it by a tangent local vector field W. We get

$$\begin{aligned} \langle R(X,Y)Z, W\rangle &= \langle -\mathbf{II}(Z,Y)D_X N + \mathbf{II}(Z,X)D_Y N, W\rangle \\ &= -\mathbf{II}(Z,Y)\langle D_X N, W\rangle + \mathbf{II}(Z,X)\langle D_Y N, W\rangle \\ &= -\mathbf{II}(Z,Y)\mathbf{II}(X,W) + \mathbf{II}(Z,X)\mathbf{II}(Y,W) \ . \end{aligned}$$

Now, if we fix v_1, v_2 a basis of T_pS and evaluate the last equation by setting $X = v_1, Y = v_2, Z = v_2, W = v_1$ and $h_{ij} = \mathbf{II}(v_i, v_j)$ we get

$$\begin{aligned} \langle R(v_1,v_2)v_2, v_1\rangle &= -\mathbf{II}(v_2,v_2)\mathbf{II}(v_1,v_1) + \mathbf{II}(v_2,v_1)\mathbf{II}(v_2,v_1) \\ &= -\left(\mathbf{II}(v_2,v_2)\mathbf{II}(v_1,v_1) - \mathbf{II}(v_1,v_2)^2\right) \\ &= -\det(h_{ij}) \ . \end{aligned} \tag{8}$$ {eq:gc5}

Now, by definition of sectional curvature and setting $g_{ij} = \mathbf{I}(v_i, v_j)$,

$$\langle R(v_1,v_2)v_2, v_1\rangle = K(v_1,v_2)\det(g_{ij}) \ . \tag{9}$$ {eq:gc6}

Here $K(v_1,v_2)$ is the sectional curvature of the plane spanned by v_1, v_2 which is simply the gaussian curvature K since we're in dimension 2.
So by putting together (8) and (9),

$$K = -\det_{\mathbf{I}}(\mathbf{II}) = -\frac{\det(h_{ij})}{\det(g_{ij})} \ .$$

Which is the same as the Gauss equation for surfaces in Euclidean space if not for a negative sign.

Let us also focus on the other equation. We can define the exterior derivative of a tensor with respect to the connection,

$$d^{\nabla}T(X,Y) = \nabla_X(TY) - \nabla_Y(TX) - T[X,Y] = (\nabla_X T)(Y) - (\nabla_Y T)(X) \ .$$

This way, Codazzi's equation becomes

$$d^{\nabla}\mathbf{II} = 0.$$

So that the two equations get the following form:

$$\begin{cases} \det_{\mathbf{I}}(\mathbf{II}) = -K \\ d^{\nabla}\mathbf{II} = 0. \end{cases} \tag{10}$$ {eq:gc}

Bonnet's theorem

The Gauss-Codazzi equations play a fundamental role when the ambient manifold has constant sectional curvature. In this case, they are not only necessary compatibility conditions for the existence of an isometric immersion, but, on a simply connected surface, they are also sufficient.

{th:bonnet}

Theorem C.1 (Bonnet)**.** *Let $(g_{ij})_{ij}$ and $(h_{ij})_{ij}$ be differentiable functions defined on an open set $V \subset \mathbb{R}^2$ to $\mathbb{R}$ such that $g = (g_{ij})_{ij}$ is is a symmetric positive definite matrix and $h = (h_{ij})_{ij}$ is symmetric. Suppose that they satisfy the equation* (10)*.*
Then for every $q \in V$ there exists a neighborhood $U \subset V$ of q and a diffeomorphism $\mathbf{x} : U \to \mathbf{x}(U) \subset \mathbb{M}^{2,1}$ such that the regular surface $\mathbf{x}(U)$ has g and h as first and second fundamental forms respectively.
Furthermore, if U is connected and if

$$\overline{\mathbf{x}} : U \to \overline{\mathbf{x}}(U) \subset \mathbb{M}^{2,1}$$

is another diffeomorphism satisfying the same conditions, then there exist an isometry $T : \mathbb{M}^{2,1} \to \mathbb{M}^{2,1}$ such that $\overline{\mathbf{x}} = T \circ \mathbf{x}$.

We redirect to [2] for a proof of the Euclidean case, whose only difference is the sign of Gauss' equation.

Mean curvature

We conclude this section by briefly recalling the definition of mean curvature, which does not differ from the usual Euclidean setting. As we have already seen, we can associate to the bilinear form $\mathbf{II}$ a linear operator $A_p : T_pS \to T_pS$, called shape operator, such that $\mathbf{II}(v,w) = \langle A_p(v), w\rangle$.
We define the mean curvature of S at p as

$$H(p) = \frac{1}{2}\operatorname{tr}(A_p) \ .$$

If we again fix a basis v_1, v_2, denote with $(g_{ij})_{ij}$ and $(h_{ij})_{ij}$ the coefficients of the first and second fundamental forms, and express A_p as $A_p(v_i) = A_i{}^k v_k$, we get

$$h_{ij} = \mathbf{II}(v_i, v_j) = \langle A_p(v_i), v_j\rangle = A_i{}^k g_{kj} \ .$$

If (g^{ij}) is the inverse matrix of (g_{ij}), it follows that

$$A_i^{\ k} = h_{ij} g^{jk},$$

and therefore

$$\mathrm{tr}(A_p) = A_i^{\ i} = h_{ij} g^{ji} = g^{ij} h_{ij} \quad .$$

So that we have

$$H = \frac{1}{2}\,\mathrm{tr}(A_p) = \frac{1}{2} A_i^{\ i} = \frac{1}{2} g^{ij} h_{ij} = \frac{1}{2}\,\mathrm{tr}_{\mathbf{I}}(\mathbf{II}) \quad .$$

Now, since

$$(g^{ij}) = \frac{1}{\det(g_{ij})} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{12} & g_{11} \end{pmatrix} \quad ,$$

we obtain

$$2H = g^{ij} h_{ij} = \frac{h_{22} g_{11} - 2h_{12} g_{12} + h_{11} g_{22}}{\det(g_{ij})} \quad .$$

As we already mentioned, there is no formal difference with surfaces immersed in $\mathbb{R}^3$.

D Riemann surfaces and harmonic maps

We here introduce some notation and some fundamental concepts of Riemann surfaces. The most crucial part will be the fact that each metric on a surface is locally conformal to the Euclidean, the relationship between a Riemannian metric and a complex structure, the Uniformization theorem and the definition of the Hopf differential.

D.1 Preliminaries on Riemann surfaces

Let us fix S a surface (a topological manifold of dimension 2).

Definition D.1. A complex atlas $\mathcal{C}$ is a collection of charts

$$\mathcal{C} = \{z_\alpha : U_\alpha \to \mathbb{C}\}$$

that covers the whole surface S and whose transition maps

$$z_\beta \circ z_\alpha^{-1} : \ z_\alpha\big(U_\alpha \cap U_\beta\big) \to z_\beta\big(U_\alpha \cap U_\beta\big)$$

are holomorphic as maps between open subsets of $\mathbb{C}$. We call $\mathscr{C}$ a complex or conformal structure if it is an atlas maximal under the subset relation. Finally we call the pair $(S, \mathscr{C})$ (just S for short) a Riemann surface.

Definition D.2. A continuous map $h : S_1 \to S_2$ between Riemann surfaces is said to be holomorphic if, in local coordinates $\{U_\alpha, z_\alpha\}$ on S_1 and $\{U'_\beta, z'_\beta\}$ on S_2,all the maps $z'_\beta \circ h \circ z_\alpha^{-1}$ are holomorphic wherever they are defined. We call f a biholomorphism if it is holomorphic, bijective and its inverse f^{-1} is holomoprhic.
We say that the two Riemann surfaces S_1 and S_2 are isomorphic if there exists a biholomoprhism between them.

Definition D.3. We say that two metrics g and g' on a surface S are conformal to each other if there exists a smooth function $\upsilon : S \to \mathbb{R}$ such that

$$g' = e^{2\upsilon} g \quad .$$

Equivalently, g' is at each point a positive multiple of g by a smooth function. The function $e^{2\upsilon}$ is called the conformal factor.

Conformal structures on 2-surfaces

Here we state the results that lead to the fact that every Riemannian surface admits a complex structure.

Theorem D.1. *Given g a Riemannian metric on S differentiable surface, and a point $p \in S$, there is a chart near p, $\varphi : U \to \mathbb{R}^2$ such that g takes the form and a smooth positive function λ such that*

$$g = \lambda^2(dx^2 + dy^2) = e^{2\upsilon}(dx^2 + dy^2),$$

where (x, y) are the coordinates defined by φ.

Definition D.4. The chart (φ, U) in the theorem is called conformal, while the coordinates (x, y) on U are called isothermal coordinates around p.

Given a conformal chart we may introduce the change of coordinates

$$\begin{cases} z = x + iy \\ \bar{z} = x - iy \end{cases} \quad \text{or equivalently} \quad \begin{cases} x = \frac{1}{2}(z + \bar{z}) \\ y = \frac{i}{2}(z - \bar{z}) \end{cases} \quad .$$

The associated coordinated fields and forms can be expressed as

$$\begin{cases} \dfrac{\partial}{\partial z} = \dfrac{1}{2}(\dfrac{\partial}{\partial x} - i\dfrac{\partial}{\partial y}) \\ \dfrac{\partial}{\partial \bar{z}} = \dfrac{1}{2}(\dfrac{\partial}{\partial x} + i\dfrac{\partial}{\partial y}) \end{cases} \quad \text{and} \quad \begin{cases} dz = dx + idy \\ d\bar{z} = dx - idy \end{cases} \quad .$$

This lets us write the Euclidean metric as

$$\begin{aligned} dx^2 + dy^2 &= dx \otimes dx + dy \otimes dy \\ &= \frac{1}{4} d(z + \bar{z}) \otimes d(z + \bar{z}) - \frac{1}{4} d(z - \bar{z}) \otimes d(z - \bar{z}) \\ &= \frac{1}{2}(dz \otimes d\bar{z} + d\bar{z} \otimes dz) \\ &= \Re(dz \otimes d\bar{z}) \\ &= dz d\bar{z} \\ &= |dz|^2 \quad , \end{aligned}$$

where the last two equalities are definitions.
So we can express the metric g as

$$g = e^{2\upsilon}|dz|^2.$$

The change of coordinates $(x, y) \mapsto z = x + iy$ yields charts with values in $\mathbb{C}$. However, this is not yet enough to conclude that S carries a complex structure, since one still has to prove that the corresponding transition maps are holomorphic. But this can be shown with a little more work:

Proposition 3. *Fix an orientation on S. Given a conformal chart (φ, U) preserving this orientation, denote by $(\tilde{\varphi}, U)$ the associated complex chart. Then*

$$\{(\tilde{\varphi}, U) \mid (\varphi, U) \text{ is an orientation-preserving conformal chart on } (S, g)\}$$

constitutes a complex structure on S.

Now, let us recall some operators that take a simple form when read through a conformal chart, for which the metric has the form $e^{2u}|dz|^2$. The proof of these formulas can be found in [7].
First, the Laplacian with respect to the metric g is

$$\Delta_g = \frac{1}{\lambda^2}(\frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}) = e^{-2u}(\frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}) = e^{-2u}\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}.$$

While the Gauss curvature of the surface takes the form

$$K_g = -\Delta_g \log(\lambda) = -2\Delta u.$$

Let us finally observe that,

$$\begin{aligned} dz^2 &= dz \otimes dz \\ &= d(x+iy) \otimes d(x+iy) \\ &= dx \otimes dx + i\,(dx \otimes dy + dy \otimes dx) - dy \otimes dy \\ &=: dx^2 - dy^2 + 2i dx dy \quad . \end{aligned}$$

Integration through a conformal chart
Given a curve γ in S, its length is defined by

$$\ell(\gamma) = \int_\gamma \lambda(z)|dz| \ .$$

On the other hand the area form can be expressed as follows

$$\lambda^2 dx \wedge dy = -\frac{i\lambda^2}{2} dz \wedge d\bar{z} \ ,$$

so that the area of a measurable set becomes

$$\begin{aligned} \text{Area}(B) &:= \int_B \lambda^2(z)\frac{i}{2} dz \wedge d\bar{z} \\ &= \int_B \lambda^2(z)\frac{i}{2} dz d\bar{z} \quad . \end{aligned}$$

D.2 Uniformization

An important property of Riemann surfaces is that they can be classified in just three categories, depending on their universal covering. This result is (a direct consequence of) the uniformization theorem. In what follows we denote by $\widehat{\mathbb{C}}$ the 2-sphere, with the metric and conformal structure induced by the Euclidean metric of $\mathbb{R}^3$.

Theorem D.2 (Uniformization Theorem)**.** *Let Σ be a simply connected Riemann surface. Then Σ is isomorphic to exactly one of the following three surfaces:*

1. *The Riemann sphere $\widehat{\mathbb{C}}$.*
2. *The complex plane $\mathbb{C}$.*
3. *The Poincaré disk $\mathbb{D}$.*

Every Riemann surface Σ admits a universal covering $\tilde{\Sigma}$, which inherits the complex structure of its base space. Since the universal covering $\tilde{\Sigma}$ of a surface Σ is simply connected, we can apply the uniformization theorem to it, making Σ fall in one of the following categories. We say that Σ is

- elliptic if $\widetilde{\Sigma} \cong \widehat{\mathbb{C}}$,
- parabolic if $\widetilde{\Sigma} \cong \mathbb{C}$,
- hyperbolic if $\widetilde{\Sigma} \cong \mathbb{D}$.

In our case the surface we consider is simply connected and noncompact (for instance because it is the graph of an entire function on $\mathbb{R}^2$) so we can exclude the elliptic case.

Corollary D.1. *If Σ is a simply connected noncompact Riemann surface, then Σ is conformally equivalent either to $\mathbb{C}$ or to $\mathbb{D}$.*

D.3 Harmonic maps into $\mathbb{H}$

An important result we need to prove is that the Gauss map of a complete space-like surface in $\mathbb{M}^{2,1}$ is harmonic if and only if the surface is a CMC cut. In order to do this we need first to give a definition of harmonicity for a map with values in $\mathbb{M}^{2,1}$.
Let us consider $f : (\mathbb{D}, e^{2\eta}(dx^2+dy^2)) \to \mathbb{M}^{2,1}$. We can define the energy density of f as

$$e(f) = e^{-2\eta}(f_x \cdot f_x + f_y \cdot f_y) \ .$$

If the map has compact support we can integrate the energy density, obtaining the energy of the map,

$$\begin{aligned} E(f) &= \int_{\mathbb{D}} e^{2\eta} e(f) dx \wedge dy \\ &= \int_{\mathbb{D}} f_x \cdot f_x + f_y \cdot f_y dx \wedge dy \quad . \end{aligned}$$

This quantity does not define an $\mathbb{R}$-valued functional on the space of all maps f, since the integral may be infinite. Nevertheless, if we consider a compactly supported deformation of f, the corresponding first variation is still well defined. This in turn lets us give the definition of a harmonic map.

Let us fix a compactly supported deformation of f, namely a smooth map

$$\xi : \mathbb{D} \times (-\varepsilon, \varepsilon) \to \mathbb{M}^{2,1}$$

such that

$$\xi(x, y, 0) = f(x, y)$$

for every $(x, y) \in \mathbb{D}$, and there exists a compact set $K \subset \mathbb{D}$ such that

$$\xi(x, y, t) = f(x, y)$$

for every $(x, y) \in \mathbb{D} \setminus K$ and every $t \in (-\varepsilon, \varepsilon)$. We call ξ of this form an admissible deformation of f.
Then, we define

$$f_t := \xi(\cdot, \cdot, t) \ .$$

Definition D.5. We say that f is a harmonic map if for every admissible deformation ξ we have

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = 0 \ .$$

The following lemma is needed to give later a characterizaion of harmonic maps.

{lm:deren1}

Lemma 1.

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\mathbb{D}} \xi_t \cdot (f_{xx} + f_{yy}) dx \wedge dy \ .$$

Proof. First we observe that

$$\begin{aligned} \left.\frac{dE(f_t)}{dt}\right|_{t=0} &= \int_{\mathbb{D}} \frac{\partial}{\partial t}\big(\xi_x \cdot \xi_x + \xi_y \cdot \xi_y\big)\Big|_{t=0} dx \wedge dy \\ &= 2 \int_{\mathbb{D}} \big(\xi_x \cdot \xi_{xt} + \xi_y \cdot \xi_{yt}\big)\big|_{t=0} dx \wedge dy \ . \end{aligned} \tag{11}$$

{eq:deren1}

Now,

$$\begin{aligned} d(\xi_t \cdot \xi_x \, dy) &= (\xi_t \cdot \xi_x)_x \, dx \wedge dy \\ &= (\xi_{xt} \cdot \xi_x + \xi_t \cdot \xi_{xx}) \, dx \wedge dy, \end{aligned}$$

so that

$$\xi_x \cdot \xi_{xt} \, dx \wedge dy = d(\xi_t \cdot \xi_x \, dy) - \xi_t \cdot \xi_{xx} \, dx \wedge dy$$

and similarly

$$\xi_y \cdot \xi_{yt} \, dx \wedge dy = -d(\xi_t \cdot \xi_y \, dx) - \xi_t \cdot \xi_{yy} \, dx \wedge dy \ .$$

Now, by substituting into (11), we obtain

$$\begin{aligned} \left.\frac{dE(f_t)}{dt}\right|_{t=0} &= -2 \int_{\mathbb{D}} \xi_t \cdot (\xi_{xx} + \xi_{yy})\big|_{t=0} dx \wedge dy \\ &\quad + 2 \int_{\mathbb{D}} d\big(\xi_t \cdot \xi_x \, dy - \xi_t \cdot \xi_y \, dx\big)\big|_{t=0} \ . \end{aligned}$$

Then, since ξ_t has compact support K, by applying Stokes' theorem we obtain that the second term of the sum in the last equation is zero. Once recognized that $\xi_{xx}|_{t=0} = f_{xx}$ and $\xi_{yy}|_{t=0} = f_{yy}$ we have proven the lemma. □

{cor:tglapnull}

Corollary D.2. *Let $f : \mathbb{D} \to \mathbb{H} \subset \mathbb{M}^{2,1}$ a smooth map. Then f is harmonic if and only if the tangent component of its Laplacian is null:*

$$(\Delta f)^T = 0 \ .$$

That is,

$$\Delta f + \langle \Delta f, f \rangle \, f = 0 \ .$$

Proof. Let us fix $\xi : \mathbb{D} \times (-\varepsilon, \varepsilon) \to \mathbb{H}$ a compactly supported deformation of f (that is, a deformation of f as above). By lemma 1,

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\mathbb{D}} \xi_t|_{t=0} \cdot \Delta f \ .$$

Being ξ a deformation of f, which is a smooth map with values in $\mathbb{H}$, we have constantly $\xi(x, y, t) \in \mathbb{H}$, hence $\xi_t(x, y)|_{t=0} \in T_{\xi(x,y,0)}\mathbb{H}^2$. So we obtain

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\mathbb{D}} \xi_t \cdot (\Delta f)^T \ . \tag{12}$$ {eq:tglaplnull1}

Now, one implication is trivial, if $(\Delta f)^T = 0$, then also the LHS is zero for every choice of ξ.

On the other hand, let us assume that

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = 0$$

for every admissible deformation and prove that $(\Delta f)^T$ is zero. We may suppose by contradiction $(\Delta f)^T \neq 0$ at some point p, this implies it is nonzero in a small neighborhood U of p.
Then we choose a smooth function $\psi \geq 0$ of compact support contained in U, such that $\psi(p) > 0$, and define the deformation

$$\xi(x, y, t) = \exp_{f(x,y)}\big(t\psi(x, y)(\Delta f)^T(x, y)\big)$$

By substituting ξ in the expression (12) we obtain

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\operatorname{supp}\psi} \psi\, |(\Delta f)^T|^2 dx \wedge dy < 0 \ ,$$

where we have the strict inequality since we have both $\psi > 0$ and $(\Delta f)^T \neq 0$ on $\operatorname{supp}\psi \subset U$. This is absurd because we showed that the leftmost side of the last expression is exactly zero.

□

Let us fix now a conformal parametrization of our surface,

$$\sigma : \mathbb{D} \to \mathbb{M}^{2,1},$$

and let

$$N : \mathbb{D} \to \mathbb{M}^{2,1}$$

be the gauss map of the surface. We want to prove the following theorem:

{th:nablh}

Theorem D.3.

$$(\Delta N)^T = 2(\nabla H) \ ,$$

where H is the mean curvature of the surface.

In order to prove it, we need a lemma.

{lm:trnabl}

Lemma 2. *Let B a 1-1 tensor on a Riemannian manifold M and $v \in T_pM$. Then*

$$\operatorname{tr}(\nabla_v B) = d(\operatorname{tr} B)(v) \ .$$

Proof. Let us suppose the dimension of the manifold to be 2 for simplicity. Let $(e_i)_i$ be an orthonormal frame around p.
Let us observe

$$\begin{aligned} d\operatorname{tr}(B)(v) &= v\left(\sum_i \langle B(e_i), e_i\rangle\right) \\ &= \sum_i \langle \nabla_v(B(e_i)), e_i\rangle + \sum_i \langle B(e_i), \nabla_v e_i\rangle \\ &= \sum_i \langle (\nabla_v B)(e_i), e_i\rangle + \sum_i \left[\langle B(\nabla_v e_i), e_i\rangle + \langle B(e_i), \nabla_v e_i\rangle\right] \\ &= \operatorname{tr}(\nabla_v B) + \sum_i \left[\langle B(\nabla_v e_i), e_i\rangle + \langle B(e_i), \nabla_v e_i\rangle\right] \ . \end{aligned} \tag{13}$$

{eq:trb1}

So that we have our thesis if we show that the second term of the expression in (13) is zero. Suppose

$$\nabla_v e_1 = a e_1 + b e_2, \qquad \nabla_v e_2 = c e_1 + d e_2 \ . \ ,$$

for certain coefficients $a, b, c, d \in \mathbb{R}$.
Observe then that

$$\begin{aligned} \langle e_i, e_j\rangle &= \delta_{ij} \\ \Rightarrow \quad v\left(\langle e_i, e_j\rangle\right) &= v(\delta_{ij}) = 0 \\ \Rightarrow \quad \langle \nabla_v e_i, e_j\rangle + \langle e_i, \nabla_v e_j\rangle &= 0 \\ \Rightarrow \quad \langle \nabla_v e_i, e_j\rangle &= -\langle e_i, \nabla_v e_j\rangle \ . \end{aligned}$$

By setting $i = j$ we obtain $\langle \nabla_v e_1, e_1\rangle = a = \langle \nabla_v e_2, e_2\rangle = d = 0$ and if $i \neq j$ we get $\langle \nabla_v e_1, e_2\rangle = b = -\langle \nabla_v e_2, e_1\rangle = -c$. Hence

$$\nabla_v e_1 = b e_2, \qquad \nabla_v e_2 = -b e_1.$$

Now we can rewrite the second term of (13),

$$\begin{aligned} &\langle B(\nabla_v e_1), e_1\rangle + \langle B(e_1), \nabla_v e_1\rangle + \langle B(\nabla_v e_2), e_2\rangle + \langle B(e_2), \nabla_v e_2\rangle = \\ &\quad b\langle B(e_2), e_1\rangle + b\langle B(e_1), e_2\rangle - b\langle B(e_1), e_2\rangle - b\langle B(e_2), e_1\rangle = 0 \ . \end{aligned}$$

Which proves the lemma. □

We are ready to prove theorem D.3.

Proof of theorem D.3. First we recall that if A is the shape operator, then $N_x = A\partial_x$ and $N_y = A\partial_y$. If we assume that $\mathbf{I} = e^{2\eta}(dx^2 + dy^2)$ then $e^{-\eta}\partial_x$, $e^{-\eta}\partial_y$ is an orthonormal frame.
By applying the covariant derivatives along ∂_x and ∂_y to N_x and N_y, respectively, we obtain

$$\begin{aligned} N_{xx}^T &= (\nabla_{\partial_x} A)(\partial_x) + A(\nabla_{\partial_x}\partial_x) \\ N_{yy}^T &= (\nabla_{\partial_y} A)(\partial_y) + A(\nabla_{\partial_y}\partial_y) \ . \end{aligned}$$

So we obtain

$$
\begin{aligned}
(\Delta_{\mathbf{I}} N)^T = e^{-2\eta}\big((\nabla_{\partial_x} A)(\partial_x) + (\nabla_{\partial_y} A)(\partial_y)\big) \\
+ e^{-2\eta} A\big(\nabla_{\partial_x}\partial_x + \nabla_{\partial_y}\partial_y\big) \quad .
\end{aligned} \tag{14}
$$

{eq:nablh1}

Now let us show that $\nabla_{\partial_x}\partial_x + \nabla_{\partial_y}\partial_y = 0$.
First, observe that since ∂_x, ∂_y are orthogonal, $\langle \partial_x, \partial_y\rangle = 0$ we have $\big\langle \nabla_{\partial_y}\partial_x, \partial_y\big\rangle = -\big\langle \partial_x, \nabla_{\partial_y}\partial_y\big\rangle$. Furthermore, from the expression of the metric we have $\langle \partial_x, \partial_x\rangle = e^{2\eta} = \langle \partial_y, \partial_y\rangle$ so that $\partial_y(\langle \partial_x, \partial_x\rangle) = \partial_x(\langle \partial_y, \partial_y\rangle)$ and also, the connection is symmetric so that $\nabla_{\partial_x}\partial_y = \nabla_{\partial_y}\partial_x$. Then,

$$
\begin{aligned}
\langle \nabla_{\partial_x}\partial_x, \partial_y\rangle &= -\langle \partial_x, \nabla_{\partial_x}\partial_y\rangle \\
&= -\frac{1}{2}\partial_y \langle \partial_x, \partial_x\rangle \\
&= -\frac{1}{2}\partial_y \langle \partial_y, \partial_y\rangle \\
&= -\big\langle \nabla_{\partial_y}\partial_y, \partial_y\big\rangle \quad ,
\end{aligned}
\qquad
\begin{aligned}
\langle \nabla_{\partial_x}\partial_x, \partial_x\rangle &= \frac{1}{2}\partial_x \langle \partial_x, \partial_x\rangle \\
&= \frac{1}{2}\partial_x \langle \partial_y, \partial_y\rangle \\
&= \langle \nabla_{\partial_x}\partial_y, \partial_y\rangle \\
&= \big\langle \nabla_{\partial_y}\partial_x, \partial_y\big\rangle \\
&= -\big\langle \partial_x, \nabla_{\partial_y}\partial_y\big\rangle \quad .
\end{aligned}
$$

At this point, by developing on the base ∂_x, ∂_y it is clear that

$$
\nabla_{\partial_x}\partial_x + \nabla_{\partial_y}\partial_y = 0 \ .
$$

So in (14) we obtain

$$
(\Delta N)^T = e^{-2\eta}\big((\nabla_{\partial_x} A)(\partial_x) + (\nabla_{\partial_y} A)(\partial_y)\big) \ . \tag{15}
$$

{eq:nablh2}

Since the shape operator A is defined by $\mathbf{II}(Y, Z) = \langle AY, Z\rangle$, we can relate the covariant derivatives of A and $\mathbf{II}$ as follows:

$$
\begin{aligned}
(\nabla_X \mathbf{II})(Y, Z) &= X(\mathbf{II}(Y, Z)) - \mathbf{II}(\nabla_X Y, Z) - \mathbf{II}(Y, \nabla_X Z) \\
&= X(\langle AY, Z\rangle) - \langle A(\nabla_X Y), Z\rangle - \langle AY, \nabla_X Z\rangle \\
&= \langle \nabla_X (AY), Z\rangle + \cancel{\langle AY, \nabla_X Z\rangle} - \langle A(\nabla_X Y), Z\rangle - \cancel{\langle AY, \nabla_X Z\rangle} \\
&= \langle (\nabla_X A)(Y), Z\rangle \quad .
\end{aligned}
$$

Then we can apply the Codazzi equation in the following form:

$$
\begin{aligned}
\big\langle (\nabla_{\partial_y} A)(\partial_y), \partial_x\big\rangle &= (\nabla_{\partial_y}\mathbf{II})(\partial_y, \partial_x) \\
&= (\nabla_{\partial_y}\mathbf{II})(\partial_x, \partial_y) \\
&= \langle (\nabla_{\partial_x} A)(\partial_y), \partial_y\rangle \ .
\end{aligned} \tag{16}
$$

{eq:nablh3}

So, finally, by taking the scalar product of (15) with ∂_x, we obtain

$$
\begin{aligned}
\big\langle (\Delta N)^T, \partial_x\big\rangle &= e^{-2\eta}\left[\langle (\nabla_{\partial_x} A)(\partial_x), \partial_x\rangle + \big\langle (\nabla_{\partial_y} A)(\partial_y), \partial_x\big\rangle\right] \\
\text{[(16)]} \qquad &= e^{-2\eta}\left[\langle (\nabla_{\partial_x} A)(\partial_x), \partial_x\rangle + \langle (\nabla_{\partial_x} A)(\partial_y), \partial_y\rangle\right] \\
&= \big\langle (\nabla_{\partial_x} A)(e^{-\eta}\partial_x), e^{-\eta}\partial_x\big\rangle + \big\langle (\nabla_{\partial_x} A)(e^{-\eta}\partial_y), e^{-\eta}\partial_y\big\rangle \\
&= \operatorname{tr}(\nabla_{\partial_x} A) \\
\text{[Lemma 2]} \qquad &= d\operatorname{tr}(A)(\partial_x) \\
&= 2\partial_x(H) = 2H_x \ .
\end{aligned}
$$

Analogously we can show that

$$\big\langle (\Delta N)^T, \partial_y \big\rangle = 2H_y \quad .$$

Therefore we have the thesis. □

We have now all the pieces needed to prove the central theorem of this subsection

{th:gaussmapharmiffconsth

Theorem D.4. *Let $\sigma : (\mathbb{D}, e^{2\eta}(dx^2 + dy^2)) \to S \subset M$ a parametrization of a space-like surface and denote with $N : \mathbb{D} \to \mathbb{M}^{2,1}$ the corresponding Gauss map. Then S has constant mean curvature if and only if N is harmonic.*

Proof. Just observe that for theorem D.3 we have

$$(\Delta N)^T = 2\nabla H.$$

Then recall from corollary D.2 that N if and only if the LHS in the last equation is zero, which in turn is equivalent to H being constant. □

D.4 Harmonic maps between Riemann surfaces

We can formulate the notion of harmonicity in a slightly more general setting. Let (Σ_1, g) and (Σ_2, h) be two Riemann surfaces, and let z and w be local conformal coordinates on Σ_1 and Σ_2, respectively, so that the metrics are written as

$$g = e^{2\eta}|dz|^2 = \lambda^2|dz|^2, \qquad h = \varrho^2(w)|dw|^2 \quad .$$

Let $f : \Sigma_1 \to \Sigma_2$ be a smooth map, and denote for simplicity by w its local expression in the chosen coordinates. In the last section we set $\Sigma_1 = \mathbb{D}$ and $\Sigma_2 = \mathbb{H}$ and defined the energy density of f as

$$e(f) = \frac{1}{\lambda^2}\big(f_x \cdot f_x + f_y \cdot f_y\big) \quad .$$

Since the metric on the target is $\varrho^2(w)|dw|^2$, we have

$$f_x \cdot f_x = \varrho^2(w)|w_x|^2, \qquad f_y \cdot f_y = \varrho^2(w)|w_y|^2,$$

and therefore

$$e(f) = \frac{\varrho^2(w)}{\lambda^2}\big(|w_x|^2 + |w_y|^2\big) \quad .$$

Now, if we introduce the complex coordinate $z = x + iy$, as before, we have $w_z = \frac{1}{2}(w_x - iw_y)$ and $w_{\bar{z}} = \frac{1}{2}(w_x + iw_y)$. Thus

$$w_x = w_z + w_{\bar{z}}, \qquad w_y = i(w_z - w_{\bar{z}}) \quad .$$

Hence

$$\begin{aligned} |w_x|^2 + |w_y|^2 &= |w_z + w_{\bar{z}}|^2 + |i(w_z - w_{\bar{z}})|^2 \\ &= |w_z + w_{\bar{z}}|^2 + |w_z - w_{\bar{z}}|^2 \\ &= 2\big(|w_z|^2 + |w_{\bar{z}}|^2\big), \end{aligned}$$

so that the energy density can be written as

$$e(f) = 2\frac{\varrho^2(w)}{\lambda^2}\big(|w_z|^2 + |w_{\bar{z}}|^2\big) \quad .$$

Then the energy of the map, when defined, becomes

$$\begin{aligned} E(f) &:= \int_{\Sigma_1} \lambda^2(z) e(f(z))\, i\, dz d\bar{z} \\ &= \int_{\Sigma_1} \varrho^2(w)\big(|w_z|^2 + |w_{\bar{z}}|^2\big)\, i\, dz d\bar{z} \quad . \end{aligned}$$

Like we already did before, we can define a harmonic map as a critical point for any admissible deformation (that is, a perturbation with compact support). The correspective Euler-Lagrange equation of such a problem is

$$w_{z\bar{z}} + 2\frac{\varrho_w}{\varrho} w_z w_{\bar{z}} = 0 \ .$$

For a complete derivation we redirect the reader to [7]. An important observation is that the local harmonic map equation does not involve the conformal factor λ of the metric on the domain. This reflects the fact that, in dimension two, harmonicity depends only on the conformal class of the metric on the source surface.

D.5 Energy densities and the Hopf differential

Energy densities and Jacobian

In addition to the energy density of the map f, it is convenient to introduce two more refined quantities, the holomorphic and anti-holomorphic energy densities. These objects will let us also express the Jacobian of f in a convenient way.

Definition D.6. We define the holomorphic and antiholomorphic energy density of a smooth map $f : \Sigma_1 \to \Sigma_2$ respectively as

$$\mathcal{H}(f) := \frac{\varrho^2(f)}{\lambda^2}|f_z|^2, \qquad \text{and} \qquad \mathcal{L}(f) := \frac{\varrho^2(f)}{\lambda^2}|f_{\bar{z}}|^2 \quad .$$

It is immediate to note that

$$e(f) = \mathcal{H}(f) + \mathcal{L}(f) \ .$$

The Jacobian of the map is by definition

$$J(f) = \frac{\varrho^2(f)}{\lambda^2}(|f_z|^2 - |f_{\bar{z}}|^2) \quad .$$

So that we also have

$$J(f) = \mathcal{H}(f) - \mathcal{L}(f) \ . \quad . \tag{17}$$ {eq:defjacob}

Quadratic differentials
To each map between Riemann surfaces we can associate a quantity that carries information about how far the map is from being conformal.

Definition D.7. Given Σ a Riemann surface we say that a symmetric 0-2 tensor field on Σ with values in $\mathbb{C}$ is compatible with the complex structure if it can be expressed locally in isothermal coordinates as $\Phi = \varphi(z)dz^2$. We say that Φ is holomorphic if φ is a holomorphic function.

It is clear that if we ha a change of coordiante $z \mapsto w = \gamma(z)$, then we can express the differentials as $dw = \gamma'(z)dz$. This means that the tensor product dz^2 will transform as

$$dw^2 = \gamma'(z)^2 dz^2.$$

As it is usual the case, a set of local representations of holomoprhic quadratic differentials $(dw_\alpha^2)_\alpha$ determines a holomorphic quadratic differential on the whole surface if they satisfy the transformation rule above.

To each map between Riemann surfaces, we can associate a quadratic differential, which gives us in some sense a characterization of harmonic local diffeomorphisms

{lm:exholdiff}

Lemma 3. *Let $f : \Sigma_1 \to \Sigma_2$ a harmonic map between Riemann surfaces, with Σ_1 being endowed with a metric of nonpositive curvature. Let z be a conformal parameter on Σ_1. Then if w is a local representation of f,*

$$q = \varrho^2(w) w_z w_{\overline{z}} dz^2$$

is a holomorphic quadratic differential on Σ_1.

Proof. Let us show first that the expression of q doesn't change under conformal transformations of the domain and the target.

First let us consider a conformal transformation on the domain, $z = \psi(z')$. Then $dz = \psi'(z')dz'$, $dz^2 = (\psi(z'))^2()dz')^2$,

$$w_{z'} = w_z \psi'(z'), \qquad \text{and} \qquad \overline{w}_{z'} = \overline{w}_z \psi'(z') \ .$$

So the metric transforms as

$$\begin{aligned} \varrho^2(w) w_{z'} \overline{w}_{z'} &= \varrho^2(w) w_z \overline{w}_z (\psi'(z'))^2 (dz')^2 \\ &= \varrho^2(w) w_z \overline{w}_z dz^2 \ . \end{aligned}$$

Hence q changes as

$$\begin{aligned} q = \varrho^2(w) w_z \overline{w}_z dz^2 &= \varrho^2(w) w_z \overline{w}_z (\psi'(z'))^2 (dz')^2 \\ &= \varrho^2(w) w_z \overline{w}_z dz^2 \ . \end{aligned}$$

Then we consider a conformal transformation of the target, $w = \phi(w')$. As before we have $dw = \phi'(w')dw'$, and

$$\begin{aligned} \varrho^2(w)|dw|^2 &= \varrho^2(\phi(w'))|\phi'(w')|^2 |dw'|^2 \\ &= \varrho^2(w')|dw'|^2 \ . \end{aligned}$$

Then

$$\begin{aligned} q &= \varrho^2(w) w_z \overline{w}_z dz^2 \\ &= \varrho^2(\phi(w'))(\phi'(w') w'_z \overline{\phi'(w')} \overline{w}_z) dz^2 \\ &= \varrho^2(\phi(w'))(|\phi'(w)|^2 w_z \overline{w}_z dz^2 \\ &= \varrho^2(w') w_z \overline{w}_z dz^2 \ . \end{aligned}$$

Finally, if f is harmonic, we may show that $\varrho^2(w) w_z \overline{w}_z$ is holomorphic

$$\begin{aligned} \frac{\partial}{\partial \overline{z}}(\varrho^2(w) w_z \overline{w}_z) &= \frac{\partial}{\partial \overline{z}}(\varrho^2(w)) w_z w_{\overline{z}} + \varrho^2(w) w_{z\overline{z}} \overline{w}_z + \varrho^2(w) w_z \overline{w}_{z\overline{z}} \\ &= 2\varrho(w)(\varrho_w(w) w_{\overline{z}} + \varrho_{\overline{w}}(w) \overline{w}_{\overline{z}}) w_z \overline{w}_z + \varrho^2(w) w_{z\overline{z}} \overline{w}_z + \varrho^2(w) w_z \overline{w}_{z\overline{z}} \\ &= \overline{w}_z(\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z) + \\ &\quad + \overline{w}_z(\varrho^2(\overline{w}) \overline{w}_{z\overline{z}} + 2\varrho(\overline{w})\varrho_{\overline{w}}(\overline{w}) \overline{w}_{\overline{z}} \overline{w}_z) \\ &= \overline{w}_z(\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z) + \\ &\quad + \overline{w}_z \overline{(\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z)} = 0 \end{aligned}$$

,

where the last equality holds by applying the local equation for harmonicity. □

We can actually give a decomposition of any symmetric 0-2 tensor field {lm:tfdec}

Lemma 4. *Let ω a symmetric 0-2 tensor field on (Σ, g) with values in $\mathbb{C}$. Then ω can be decomposed uniquely as*

$$\omega = \omega^{(2,0)} + \omega^{(1,1)} + \omega^{(0,2)},$$

where $\omega^{(2,0)}$ and $\overline{\omega^{(0,2)}}$ are quadratic differentials compatible with the complex structure and $\omega^{(1,1)}$ is a multiple of the metric g.

Proof. Just observe that once fixed local coordinates u, v the symmetric forms $dz^2 = du^2 - dv^2 + 2idudv$, $d\bar{z} = du^2 - dv^2 - 2idudv$ and $|dz|^2 = du^2 + dv^2$ are three linearly independent elements in a three-dimensional space, hence a base. □

We can use this lemma to find a decomposition of the pull-back metric $(f^*(h))$. What we find is actually the quadratic differential we defined in lemma 3

Lemma 5. *Let $f : (\Sigma_1, g) \to (\Sigma_2, h)$ a map two Riemann surfaces with Σ_1 being hyperbolic. Let q be the quadratic differential associated to f as in lemma 3. Then*

$$q = (f^*(h))^{(2,0)} \ .$$

Moreover we have that

$$f^*(h) = q + eg + \bar{q} \ .$$

Proof. Let us assume as usual that $g = \lambda^2 |dz|^2$ and $h = \varrho^2 |dw|^2$ are the metrics respectively on Σ_1 and Σ_2, and that $w = w(z)$ is a local representation of the map f. Then we have

$$f^*(dw) = dw = w_z dz + w_{\overline{z}} d\overline{z} \qquad \text{and} \qquad f^*(d\overline{w}) = \overline{w}_z dz + \overline{w}_{\overline{z}} d\overline{z} \ .$$

Now, if we pull back h we obtain

$$\begin{aligned} f^*(h) &= \varrho^2(w)|f^*(dw)|^2 \\ &= \varrho^2(w)(w_z dz + w_{\overline{z}} d\overline{z})(\overline{w}_z dz + \overline{w}_{\overline{z}} d\overline{z}) \\ &= \varrho^2(w)\big(w_z \overline{w}_z dz^2 + (|w_z|^2 + |w_{\overline{z}}|^2)|dz|^2 + \overline{w}_{\overline{z}} w_{\overline{z}} d\overline{z}^2\big) \ . \end{aligned}$$

Hence proving the decomposition. □

This motivates the definition of the Hopf differential

Definition D.8. Given $f : (\sigma_1, g) \to (\Sigma_2, h)$ between two Riemann surfaces, we define the Hopf differential as the quadratic differential

$$\mathrm{Hopf}(f) = (f^*(h))^{(2,0)} \ .$$

This lets us state an important result, that is a partial inverse of lemma 3.

{cor:diffeopresthenharm}

Corollary D.3. *If f is a local diffeomorphism, so that it is orientation-preserving, then if* $\mathrm{Hopf}(f)$ *is holomorphic, f is harmonic.*

Proof. In the proof of lemma 3 we showed that

$$\begin{aligned} \frac{\partial}{\partial \overline{z}}(\varrho^2(w) w_z \overline{w}_z) = \overline{w}_z\big(\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z\big) \\ + w_z \overline{\big(\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z\big)}. \end{aligned}$$

Since $\mathrm{Hopf}(f)$ is holomorphic, the LHS is zero. Then set

$$F := \varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z.$$

It is clear that F is zero where f is harmonic. Then we observe

$$\begin{aligned} 0 &= \frac{\partial}{\partial \overline{z}}(\varrho^2(w) w_z \overline{w}_z) \\ &\Rightarrow \overline{w}_z F + w_z \overline{F} = 0 \\ &\Rightarrow \overline{w_{\overline{z}}} F = -w_z \overline{F} \\ &\Rightarrow |w_{\overline{z}}|^2 |F|^2 = |w_z|^2 |F|^2 \\ &\Rightarrow \big(|w_z|^2 - |w_{\overline{z}}|^2\big)|F|^2 = 0. \end{aligned}$$

Since f is orientation-preserving local diffeomorphism,

$$|w_z|^2 - |w_{\overline{z}}|^2 = J(f) > 0,$$

hence $F = 0$ and f is harmonic. □

We can calculate, pointwise, a norm for a quadratic differential. Fix $(e_i)_i$ a local orthonormal frame on (Σ_1, g). Let us define the g norm of the quadratic differential q as

$$\|q\|_g = \frac{1}{4} \sum_{i,j} |q(e_i, e_j)|^2 \ .$$

It is possible to rewrite this quantity in terms of the energies of the map f. If the metric is $\lambda^2|dz|^2$, two orthonromal vectors are $e_1 = \frac{1}{\lambda}\frac{\partial}{\partial x}$ and $e_2 = \frac{1}{\lambda}\frac{\partial}{\partial y}$, hence

$$dz^2(e_1, e_1) = \frac{1}{\lambda^2},\ dz^2(e_1, e_2) = i\frac{1}{\lambda^2},\ dz^2(e_2, e_2) = -\frac{1}{\lambda^2}\ .$$

So that we have

$$\begin{aligned}
\|q\|_g^2 &= \frac{1}{4}\sum_{ij}|q(e_i, e_j)|^2 \\
&= \frac{1}{4}\left(|q(e_1, e_1)|^2 + 2|q(e_1, e_2)|^2 + |q(e_2, e_2)|^2\right) \\
&= \frac{1}{4}|q(z)|^2\left(|dz(e_1, e_1)|^2 + 2|q(e_1, e_2)|^2 + |q(e_2, e_2)|^2\right) \\
&= \frac{|q(z)|^2}{\lambda^4(z)}\ .
\end{aligned}$$

If we apply calculate the norm of the Hopf differential of a smooth $f : \Sigma_1 \to \Sigma_2$, with local representation $w = w(z)$,

$$\|\operatorname{Hopf}(f)\|_g^2(z) = \frac{\varrho^4(w)}{\lambda^4}|w_z|^2|w_{\overline{z}}|^2\ .$$

By recognizing the terms in this last equation we have immediately the following lemma.

The second fundamental form and the Hopf differential of the Gauss map

We now present some observations that will illustrate how the Hopf differential of the Gauss map is related to the second fundamental form.

{lm:lemmaqforms}

Lemma 6. *Let x, y be some coordinates. Then*

$$a(dx^2 - dy^2) + 2bdxdy = \frac{1}{2}[(a - ib)(dx + idy)^2 + (a + ib)(dx - idy)^2]$$

Proof. By calculation, (remember the shorthand $dxdy = \frac{1}{2}(dx \otimes dy + dy \otimes dx),\ dx^2 = dx \otimes dx,\ dy^2 = dy \otimes dy$)

$$\begin{aligned}
(a - ib)(dx + idy)^2 &= (a - ib)(dx^2 - dy^2 + 2idxdy) \\
(a + ib)(dx - idy)^2 &= (a + ib)(dx^2 - dy^2 - 2idxdy)\ ,
\end{aligned}$$

so that

$$\begin{aligned}
(a - ib)(dx + idy)^2 + (a + ib)(dx - idy)^2 &= \\
&= adx^2 - ady^2 + 2aidxdy - ibdx^2 + ibdy^2 + 2bdxdy \\
&+ adx^2 - ady^2 - 2aidxdy + ibdx^2 - ibdy^2 + 2bdxdy = \\
&= 2a(dx^2 - dy^2) + 4bdxdy\ .
\end{aligned}$$

□

Hence if we set $z = x + iy, \bar{z} = x - iy$ we obtain

$$a(dx^2 - dy^2) + 2bdxdy = \frac{1}{2}\left[(a - ib)dz^2 + (a + ib)d\bar{z}^2\right] .$$

Let us now take α a symmetric 2-form on S. Then there is a morphism of the tangent bundle $L : TM \to TM$ such that

$$\alpha(\cdot, \cdot) = \mathbf{I}(L\cdot, \cdot) .$$

Which is self-adjoint, since α is symmetric by hypothesis. L is at every point a linear endomorphism, and we can decompose it using $\mathbf{I}$ in a traceless part and a diagonal part,

$$L = \frac{\operatorname{tr} L}{2} \operatorname{Id} + L_0 .$$

Now, let us suppose that (x, y) are isothermal coordinates. We can write L_0 as

$$\begin{aligned} L_0(\partial_x) &= a\partial_x + b\partial_y \\ L_0(\partial_y) &= c\partial_x + d\partial_y , \end{aligned}$$

and since L_0 is symmetric and traceless, we have $d = -a$ and $c = b$. Hence we can represent L_0 with the matrix

$$[L_0] = \begin{bmatrix} a & b \\ b & -a \end{bmatrix} .$$

Now, as in lemma 4 we may suppose to write α as

$$\alpha = \alpha^{(2,0)} + \alpha^{(1,1)} + \alpha^{(0,2)} . \tag{18}$$ {eq:dec}

We may observe that, in the decomposition (18), from the theorem we have that $\alpha^{(2,0)}$ and $\alpha^{(0,2)}$ are locally multiples of dz^2 and $d\bar{z}^2$ respectively, which are traceless. Hence we observe

$$\begin{aligned} \alpha = \alpha^{(2,0)} + \alpha^{(1,1)} + \alpha^{(0,2)} &= \mathbf{I}(L\cdot, \cdot) \\ &= \mathbf{I}\left(\left(\tfrac{1}{2}\operatorname{tr} L \operatorname{Id} + L_0\right)\cdot, \cdot\right) \\ &= \tfrac{1}{2}\operatorname{tr}(L)\mathbf{I}(\cdot, \cdot) + \mathbf{I}(L_0\cdot, \cdot) . \end{aligned}$$

By separating the traceless part and the trace part we have

$$\alpha^{(1,1)} = \tfrac{1}{2}\operatorname{tr}(L)\mathbf{I}, \qquad \alpha^{(2,0)} + \alpha^{(0,2)} = \mathbf{I}(L_0\cdot, \cdot) . \tag{19}$$ {eq:trthm3}

Also, again from 4, since α is real valued, we have $\alpha^{(0,2)} = \overline{\alpha^{(2,0)}}$. We may now prove the following important lemma

{lm:hopfdifflocalform}

Lemma 7. *With the notation above, it holds that*

$$\alpha^{(2,0)} = \tfrac{1}{2}(a - ib)dz^2 .$$

Proof. Let us fix $v = v_x\partial_x + v_y\partial_y$, by looking at the associated quadratic form to $\mathbf{I}(L_0\cdot, \cdot)$,

$$\begin{aligned} \mathbf{I}(L_0(v_x\partial_x + v_y\partial_y), v_x\partial x + v_y\partial_y) &= \mathbf{I}(v_xL_0\partial_x + v_yL_0\partial_y, v_x\partial x + v_y\partial_y) \\ &= \mathbf{I}(v_x(a\partial_x + b\partial_y) + v_y(b\partial_x - a\partial_y), v_x\partial x + v_y\partial_y) \\ &= av_x^2 + bv_xv_y + bv_xv_y - av_y^2 . \end{aligned}$$

we have

$$\begin{aligned} \mathbf{I}(L_0\cdot,\cdot) &= a(dx^2-dy^2)+2bdxdy \\ \text{[Lemma 6]} \quad &= \tfrac{1}{2}(a-ib)(dx+idy)^2+\tfrac{1}{2}(a+ib)(dx-idy)^2 \ . \end{aligned}$$

Therefore

$$\alpha^{(2,0)} = \tfrac{1}{2}(a-ib)dz^2 \ , \tag{20}$$ {eq:trthm4}

as we intended. □

Let now α be the second fundamental form $\mathbf{II}$, so that L is the shape operator A. We consider the Gauss map $N:\mathbb{D}\to\mathbb{H}$ and the pull-back of the metric of $\mathbb{H}$ through N, which we can write as

$$\begin{aligned} N^*(g_{\mathbb{H}^2})(v,w) &= \langle D_vN, D_wN\rangle \\ \text{[def. of } A\text{]} \quad &= \mathbf{I}(A(v),A(w)) \\ [A \text{ is self-adjoint}] \quad &= \mathbf{I}(A^2v,w) \ . \end{aligned} \tag{21}$$ {eq:trthm2}

We can apply Cayley-Hamilton's theorem on $p_A(x)=x^2-\mathrm{tr}(A)x+\det(A)$, obtaining

$$p_A(A)=0 \Rightarrow A^2=\mathrm{tr}(A)A-\det(A)\mathbb{1} \ .$$

Recalling that $\mathrm{tr}(A)=2H$ and $\det(A)=-K$, we obtain

$$\begin{aligned} A^2 &= 2HA+K\,\mathrm{Id} \\ &= 2H(A_0+H\,\mathrm{Id})+K\,\mathrm{Id} \\ &= 2HA_0+(2H^2+K)\,\mathrm{Id} \ . \end{aligned} \tag{22}$$ {eq:trthm1}

Which implies in particular that $e=2H^2+K$. Now, the relations (21) and (22) along with the general (19) give us

$$N^*(g_{\mathbb{H}})^{(2,0)}+\overline{N^*(g_{\mathbb{H}})^{(2,0)}} = 2H\,\mathbf{I}(A_0\cdot,\cdot) \ .$$

So that by applying the identity (20) to $\alpha=2H\,\mathbf{II}_0$ we obtain

$$N^*(g_{\mathbb{H}})^{(2,0)} = H(a-ib)dz^2 \ .$$

And we can rewrite

$$\mathrm{Hopf}(N)+\overline{\mathrm{Hopf}(N)} = 2H\,\mathbf{I}(A_0\cdot,\cdot) \ .$$

As usual let us write $\mathbf{II}$ locally as

$$\mathbf{II}=h_{ij}dx^idx^j, \quad \text{and suppose} \quad \mathrm{Hopf}(N)=\varphi\,dz^2.$$

Since $H=1$ and $\mathbf{I}=e^{2w}(dx^2+dy^2)$, we have

$$h_{11}+h_{22}=2e^{2w}.$$

Moreover, from

$$2\mathbf{II}_0=\mathrm{Hopf}(N)+\overline{\mathrm{Hopf}(N)}=\varphi\,dz^2+\overline{\varphi}\,d\bar{z}^2$$

and Lemma 6, writing $\varphi = a - ib$, we obtain

$$\begin{aligned} 2\mathbf{II}_0 &= 2a(dx^2 - dy^2) + 4bdxdy \\ &= (h_{11} - h_{22})(dx^2 - dy^2) + 4h_{12}dxdy, \end{aligned}$$

hence

$$h_{11} - h_{22} = 2\Re(\varphi), \qquad h_{12} = -\Im(\varphi).$$

We note that these relations can be used to reconstruct the differential φdz^2, by solving the linear

$$\begin{cases} h_{11} + h_{22} = 2e^{2w} \\ h_{11} - h_{22} = 2\Re(\varphi) \\ h_{12} = -\Im(\varphi) \end{cases} . \tag{23}$$ {eq:gausspde1}

The Codazzi equation is equivalent to Φ holomorphic
The relations obtained above connect the second fundamental form with the Hopf differential of the Gauss map. In particular, when the mean curvature is constant, they allow us to prove that the Codazzi equation for the second fundamental form is equivalent to the holomorphicity of the Hopf differential of the Gauss map.

First we need to remind that from the compatibility of ∇ with the metric we get $(\nabla_X \mathbf{I})(Y, Z) = 0$, so that

$$\begin{aligned} (\nabla_X \mathbf{II})(Y, Z) &= (\nabla_X (H\mathbf{I} + \mathbf{II}_0))(Y, Z) \\ [H \text{ is constant, } \nabla\mathbf{I} = 0] \qquad &= (\nabla_X \mathbf{II}_0)(Y, Z) \ . \end{aligned}$$

Hence the Codazzi equation can be rewritten exclusively in terms of the traceless part

$$(\nabla_X \mathbf{II})(Y, Z) = (\nabla_Z \mathbf{II})(X, Y) \quad \Leftrightarrow \quad (\nabla_X \mathbf{II}_0)(Y, Z) = (\nabla_Z \mathbf{II}_0)(X, Y) \ .$$

Let us now choose local isothermal coordinates (x, y). Since A_0 is self-adjoint and traceless, with respect to the frame (∂_x, ∂_y) it is represented by a matrix of the form

$$[A_0] = \begin{pmatrix} a & b \\ b & -a \end{pmatrix} \ .$$

And thanks to Lemma 7, we can write the Hopf differential of N as

$$\Phi = \tfrac{1}{2}(a - ib)dz^2 \ .$$

We then recall the definition of the derivative of a tensor,

$$(\nabla_X \mathbf{II}_0)(Y, Z) = X(\mathbf{II}_0(Y, Z)) - \mathbf{II}_0(\nabla_X Y, Z) - \mathbf{II}_0(Y, \nabla_X Z) \ ,$$

so that we may explicit the equation in the basis. In order to do that we need the Christoffel symbols for the metric $g = e^{2w}(dx^2 + dy^2)$.
Let Γ^k_{ij} be such that

$$\nabla_{\partial_i} \partial_j = \Gamma^k_{ij} \partial_k \ .$$

Since $\langle \partial_x, \partial_x \rangle = \langle \partial_y, \partial_y \rangle = e^{2w}$ and $\langle \partial_x, \partial_y \rangle = 0$, the compatibility with the metric yields

$$\begin{aligned}
\Gamma^1_{11} e^{2w} &= \langle \nabla_{\partial_x} \partial_x, \partial_x \rangle = \frac{1}{2} \partial_x \langle \partial_x, \partial_x \rangle = e^{2w} w_x \\
\Gamma^2_{11} e^{2w} &= \langle \nabla_{\partial_x} \partial_x, \partial_y \rangle = -\frac{1}{2} \partial_y \langle \partial_x, \partial_x \rangle = -e^{2w} w_y,
\end{aligned}$$

hence $\Gamma^1_{11} = w_x$ and $\Gamma^2_{11} = -w_y$. Similarly,

$$\begin{aligned}
\Gamma^1_{12} e^{2w} &= \langle \nabla_{\partial_x} \partial_y, \partial_x \rangle = \frac{1}{2} \partial_y \langle \partial_x, \partial_x \rangle = e^{2w} w_y, \\
\Gamma^2_{12} e^{2w} &= \langle \nabla_{\partial_x} \partial_y, \partial_y \rangle = \frac{1}{2} \partial_x \langle \partial_y, \partial_y \rangle = e^{2w} w_x.
\end{aligned}$$

so $\Gamma^1_{12} = w_y$ and $\Gamma^2_{12} = w_x$. Since the connection is torsion free, $\Gamma^k_{21} = \Gamma^k_{12}$. Finally,

$$\begin{aligned}
\Gamma^1_{22} e^{2w} &= \langle \nabla_{\partial_y} \partial_y, \partial_x \rangle = -\frac{1}{2} \partial_x \langle \partial_y, \partial_y \rangle = -e^{2w} w_x, \\
\Gamma^2_{22} e^{2w} &= \langle \nabla_{\partial_y} \partial_y, \partial_y \rangle = \frac{1}{2} \partial_y \langle \partial_y, \partial_y \rangle = e^{2w} w_y.
\end{aligned}$$

so $\Gamma^1_{22} = -w_x$ and $\Gamma^2_{22} = w_y$. Therefore

$$\begin{aligned}
&\nabla_{\partial_x} \partial_x = w_x \partial_x - w_y \partial_y \ , \\
&\nabla_{\partial_x} \partial_y = w_y \partial_x + w_x \partial_y = \nabla_{\partial_y} \partial_x \ , \\
&\nabla_{\partial_y} \partial_y = -w_x \partial_x + w_y \partial_y \ .
\end{aligned}$$

We observe that $\mathbf{II}_0$ is symmetric, and therefore $\nabla \mathbf{II}_0$ is symmetric in the last two entries, therefore the Codazzi equations in the basis (∂_x, ∂_y) reduce to

$$\begin{cases}
(\nabla_{\partial_x} \mathbf{II}_0)(\partial_x, \partial_y) &= (\nabla_{\partial_y} \mathbf{II}_0)(\partial_x, \partial_x), \\
(\nabla_{\partial_x} \mathbf{II}_0)(\partial_y, \partial_y) &= (\nabla_{\partial_y} \mathbf{II}_0)(\partial_x, \partial_y).
\end{cases}$$

We can calculate $(\nabla_{\partial_x} \mathbf{II}_0)(\partial_x, \partial_y)$ and $(\nabla_{\partial_y} \mathbf{II}_0)(\partial_x, \partial_x)$,

$$\begin{aligned}
(\nabla_{\partial_x} \mathbf{II}_0)(\partial_y, \partial_x) &= \partial_x \big(\mathbf{II}_0(\partial_y, \partial_x) \big) - \mathbf{II}_0(\nabla_{\partial_x} \partial_y, \partial_x) - \mathbf{II}_0(\partial_y, \nabla_{\partial_x} \partial_x) \\
&= \partial_x b - \mathbf{II}_0(w_y \partial_x + w_x \partial_y, \partial_x) - \mathbf{II}_0(\partial_y, w_x \partial_x - w_y \partial_y) \\
&= b_x - (w_y a + w_x b) - (w_x b + w_y a) \\
&= b_x - 2 w_x b - 2 w_y a, \\
(\nabla_{\partial_y} \mathbf{II}_0)(\partial_x, \partial_x) &= \partial_y \big(\mathbf{II}_0(\partial_x, \partial_x) \big) - \mathbf{II}_0(\nabla_{\partial_y} \partial_x, \partial_x) - \mathbf{II}_0(\partial_x, \nabla_{\partial_y} \partial_x) \\
&= \partial_y a - \mathbf{II}_0(w_y \partial_x + w_x \partial_y, \partial_x) - \mathbf{II}_0(\partial_x, w_y \partial_x + w_x \partial_y) \\
&= a_y - (w_y a + w_x b) - (w_y a + w_x b) \\
&= a_y - 2 w_y a - 2 w_x b.
\end{aligned}$$

Hence

$$(\nabla_{\partial_x} \mathbf{II}_0)(\partial_x, \partial_y) = (\nabla_{\partial_y} \mathbf{II}_0)(\partial_x, \partial_x) \Leftrightarrow b_x = a_y \ .$$

Developing the calculations in the same way we obtain the equivalence

$$(\nabla_{\partial_x}\mathbf{II}_0)(\partial_y,\partial_y) = (\nabla_{\partial_y}\mathbf{II}_0)(\partial_x,\partial_y) \Leftrightarrow b_y = -a_x \ .$$

So that the Codazzi equations reduce to

$$\begin{cases} a_x = (-b)_y \\ a_y = -(-b)_x \end{cases} ,$$

which are the Cauchy-Riemann equations for $\varphi = a - ib$. We can summarize the content of this section in a statement of a theorem

{th:codhophol}

Theorem D.5. *Let S be a constant mean curvature surface, and let $N : \mathbb{D} \to \mathbb{H}$ be its Gauss map. Then the Hopf differential of N is holomorphic if and only if the second fundamental form $\mathbf{II}$ of S satisfies the Codazzi equations.*

D.6 A different form for Gauss' equation

{sc:geq}

We shall now prove the equivalence of Gauss' equation with a certain PDE on $\mathbb{D}$, which we will derive with respect to the Euclidean backgroun metric $|dz|^2$ and then rewrite it with respect to the hyperbolic metric.

Let $\mathbf{I} = e^{2w}|dz|^2$, and let (x, y) be the corresponding isothermal coordinates, with associated complex coordinate $z = x + iy$. It is clear that

$$\det(g_{ij}) = e^{4w} \ .$$

Let moreover

$$\mathbf{II} = h_{ij}\, dx^i dx^j \quad \text{and} \quad \Phi = \mathrm{Hopf}(N) = \varphi\, dz^2,$$

where N is the Gauss map of the immersion $S \hookrightarrow \mathbb{M}^{2,1}$. Recall Gauss' equation:

$$K = -\det_{\mathbf{I}}(\mathbf{II}) = -\frac{\det(h_{ij})}{\det(g_{ij})} \ .$$

We already know the expression of $\det(g_{ij})$, we need to rewrite $\det(h_{ij})$.

$$\begin{aligned} \det(h_{ij}) &= h_{11}h_{22} - h_{12}^2 \\ &= \tfrac{1}{4}\left[(h_{11}+h_{22})^2 - (h_{11}-h_{22})^2\right] - h_{12}^2 \\ \text{[(23)]} \quad &= \tfrac{1}{4}\left[(2e^{2w})^2 - (2\Re(\varphi))^2\right] - \Im(\varphi)^2 \\ &= e^{4w} - \Re(\varphi)^2 - \Im(\varphi)^2 \\ &= e^{4w} - |\varphi|^2 \ . \end{aligned}$$

Now we recall that

$$K = -e^{-2w}\Delta w \ .$$

So, substituting in Gauss' equation, we obtain

$$\begin{aligned} -e^{-2w}\Delta w &= -\frac{e^{4w} - |\varphi|^2}{e^{4w}} \\ &= -1 + |\varphi|^2 e^{-4w} \ , \end{aligned}$$

and therefore

$$\Delta w = e^{2w} - |\varphi|^2 e^{-2w} \quad . \tag{24}$$ {eq:gausspde2}

This is the Euclidean form of the equation, which will be studied in the later sections.

For completeness, let us now rewrite the same equation with respect to the hyperbolic background metric

$$ds_p^2 = \frac{4}{(1-|z|^2)^2}|dz|^2.$$

If the first fundamental form is expressed by

$$\mathbf{I} = e^{2\tilde{w}} ds_p^2,$$

then

$$e^{2w}|dz|^2 = e^{2\tilde{w}} \frac{4}{(1-|z|^2)^2}|dz|^2,$$

hence

$$w = \tilde{w} + \log\left(\frac{2}{1-|z|^2}\right) \quad . \tag{25}$$ {eq:gausspde3}

Moreover,

$$\Delta_p = \frac{(1-|z|^2)^2}{4}\Delta$$

and

$$\Delta \log\left(\frac{2}{1-|z|^2}\right) = \frac{4}{(1-|z|^2)^2}.$$

Therefore

$$\begin{aligned}
\Delta w &= \Delta\tilde{w} + \Delta \log\left(\frac{2}{1-|z|^2}\right) \\
&= \frac{4}{(1-|z|^2)^2}\Delta_p\tilde{w} + \frac{4}{(1-|z|^2)^2} \\
&= \frac{4}{(1-|z|^2)^2}\left(\Delta_p\tilde{w} + 1\right) \quad .
\end{aligned}$$

On the other hand, by (25),

$$e^{2w} = \frac{4}{(1-|z|^2)^2}e^{2\tilde{w}}.$$

Also, if $\|\Phi\|$ denotes the norm of Φ with respect to the Poincaré metric ds_p^2, then

$$\|\Phi\|^2 = \frac{(1-|z|^2)^4}{4^2}|\varphi|^2,$$

that is,

$$|\varphi|^2 = \frac{16}{(1-|z|^2)^4}\|\Phi\|^2.$$

Hence

$$|\varphi|^2 e^{-2w} = \frac{4}{(1-|z|^2)^2}\|\Phi\|^2 e^{-2\tilde{w}}.$$

Substituting these expressions into (24), we get

$$\frac{4}{(1-|z|^2)^2}\left(\Delta_p \tilde{w} + 1\right) = \frac{4}{(1-|z|^2)^2} e^{2\tilde{w}} - \frac{4}{(1-|z|^2)^2}\|\Phi\|^2 e^{-2\tilde{w}} \quad ,$$

and therefore

$$\Delta_p \tilde{w} = e^{2\tilde{w}} - \|\Phi\|^2 e^{-2\tilde{w}} - 1 \quad . \tag{26}$$ {eq:gausspde4}

This is exactly the form of the PDE found in [10]. Again, it might be useful to summarize this result in a theorem statement.

{thm:gausseqnthm}

Theorem D.6. *Let S be a constant mean curvature 1 surface, let $N : \mathbb{D} \to \mathbb{H}$ be its Gauss map, and let $\Phi = \phi\, dz^2$ be its Hopf differential. If the induced metric on S is written as*

$$\mathbf{I} = e^{2w} ds_p^2,$$

then the Gauss equation for S is equivalent to

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1.$$

E CMC surfaces and harmonic maps

We now introduce the class of surfaces that will be studied throughout this section.

Definition E.1. A constant mean curvature cut is a complete entire spacelike surface $S \subset \mathbb{M}^{2,1}$ with constant mean curvature $H > 0$.

This is essentially the same setting we exposed in Section B if not for the additional condition of having a constant mean curvature. The first step for the desired classification is the following proposition, which we proved in Section B if not for the last result, which can be found in [4]

{pr:convfun}

Proposition 4. *Let S be a complete spacelike surface in $\mathbb{M}^{2,1}$ with constant mean curvature H. Then S can be represented as the graph of an entire function $x^3 = f(x^1, x^2)$. Moreover, f is convex, and therefore the Gaussian curvature of S satisfies $K \leq 0$. More precisely, either $K < 0$ everywhere or if $K = 0$ at some point then $K \equiv 0$ and the surface is parabolic.*

An immediate consequence of being the graph of a convex function is that the second fundamental form of the surface is positive definite, from which we get that the mean curvature must be positive.

We observe now that we may suppose the mean curvature of S to be 1. In fact, if $\iota : S \hookrightarrow \mathbb{M}^{2,1}$ is an immersion with constant mean curvature $H = k > 0$, then we may consider the dilated immersion

$$\tilde{\iota} = \mu\iota,$$

which has constant mean curvature $\tilde{H} = \mu^{-1}k$. Choosing $\mu = k$, we obtain a surface with constant mean curvature 1.

Therefore, there exists a bijection between complete space-like surfaces with constant mean curvature k and complete space-like surfaces with constant mean curvature 1. Hence a classification of the latter immediately yields a classification of the former. We may then assume from now on that the surfaces have constant mean curvature 1.

Let us then fix S a hyperbolic CMC cut with mean curvature 1. The hyperbolic hypothesis means that using the uniformization theorem we can find a conformal diffeomorphism

$$\Lambda : \mathbb{D} \to S.$$

Let us now consider the Gauss map of S, $u : S \to \mathbb{H}$. By composing with Λ read it as a map

$$u : \mathbb{D} \to \mathbb{H} \subset \mathbb{M}^{2,1}.$$

The hypothesis on the mean curvature of S being constant lets us apply the Theorem D.4 and obtain that u is harmonic.

Now, we can pull back the metric of S through the conformal parametrization and express it in terms of u:

{prop:indmetr}

Proposition 5. *Suppose that S has constant mean curvature 1. Then the pulled-back induced metric $g = \Lambda^*(\mathbf{I})$ of S and the respective Gaussian curvature are given by*

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2, \qquad K = \frac{|u_{\overline{z}}|^2}{|u_z|^2} - 1 \quad .$$

Proof. We redirect the reader to [1]. □

We now need a claim we will prove later in the section.

{cl:jhupos}

Claim 1. The Gauss map u satisfies

$$J(u) > 0 \qquad \text{and} \qquad \mathcal{H}(u) > 0 \ .$$

The fact that $J(u) > 0$ along with u being harmonic lets us apply Corollary D.3 and conclude that its Hopf differential is a holomorphic quadratic differential. So, if we denote by

$$\mathrm{Hopf}(u)(z) = \varphi(z)dz^2$$

then $\varphi(z) = \varrho^2(u(z))u_z\overline{u}_{\overline{z}}$ is holomorphic. For brevity we set $\Phi := \mathrm{Hopf}(u)$

Let us now prove the claim

Proof of Claim 1. First, by Proposition 5 we have that the metric $g := \Lambda^*(\mathbf{I}_S)$ is expressed by

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2 \ .$$

Then we recall that the metric on $\mathbb{D}$ at a point v is $ds_p^2|_v := \frac{4}{(1-|v|^2)^2}|dz|^2$. So we have

$$\begin{aligned} g &= \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2 \\ &= ds_p^2|_u|u_z|^2 \\ &= ds_p^2|_u\mathcal{H}(u) \quad . \end{aligned}$$

We know that both g and ds_p^2 are Riemannian metrics, so that we obtain $\mathcal{H}(u) > 0$.

Now, from the second equation of Proposition 5 we have

$$K = \frac{|u_{\overline{z}}|^2}{|u_z|^2} - 1$$

so that

$$\begin{aligned} |u_z|^2 K &= |u_{\overline{z}}|^2 - |u_z|^2 \\ &= -J(u) \quad . \end{aligned}$$

Then

$$J(u) = -|u_z|^2 K > 0 \ ,$$

since $K < 0$, which is true by applying Proposition 4 and recalling that S is hyperbolic. □

From the proof we can also observe that the metric induced on S can be expressed as $g = \mathcal{H}ds_p^2 = e^{2w}ds_p^2$.

We have proven the following theorem:

{th:pdenecthm}

Theorem E.1. *Let S be a hyperbolic constant mean curvature cut with mean curvature equal to 1, let $u : \mathbb{D} \to \mathbb{H}$ be the Gauss map of S read through the conformal parametrization Λ and $\Phi = \varphi(z)dz^2$ its Hopf differential. Then*

- *Φ is holomorphic as a quadratic differential,*
- *u preserves the orientation,*
- *$w := \frac{1}{2}\log\mathcal{H}(u)$ is a well defined function on $\mathbb{D}$,*
- *$e^{2w}ds_p^2$ is the metric of S read through the conformal parametrization Λ,*

F Classification of hyperbolic CMC cuts

In the last section we essentially followed the sequence of constructions

$$S \longmapsto u \longmapsto (\Phi, w) \longmapsto \Phi, \tag{27}$$

{eq:infrel}

where S is a hyperbolic CMC-1 cut in $\mathbb{M}^{2,1}$; $u : \mathbb{D} \to \mathbb{H}$ is the Gauss map of S read through the conformal parametrization obtained by the uniformization theorem; Φ is the Hopf differential of u and $w = \frac{1}{2}\log\mathcal{H}(u)$ with $\mathcal{H}$ denoting the holomorphic energy of u. Furthermore we showed that the involved quantities

satisfy the properties in Theorem E.1

As it is, the assignment from the surface S to the holomorphic quadratic differential on $\mathbb{D}$, $S \mapsto \Phi$, is not well defined, because it depends on the choice of the conformal parametrization. This ambiguity disappears after passing to equivalent classes by considering the map

$$\mathscr{A} : \mathbf{E} \to \mathbf{Q}_{\mathbb{D}} \quad ,$$

where $\mathbf{E}$ is the space of surfaces S up to isometry and $\mathbf{Q}_{\mathbb{D}}$ is the space of holomorphic quadratic differentials up to change of conformal coordinates. This is in fact justified because we can't distinguish from an intrinsic point of view two isometric surfaces, while a quadratic differential compatible with the complex structure should be intended independently from its local representation.
So we can prove the theorem:

{th:class}

Theorem F.1. *The map $\mathscr{A}$ is well defined and bijective.*

The proof of this theorem requires an existence and uniqueness result for the conformal factor of the metric induced on S, expressed in terms of a holomorphic quadratic differential which will later be identified with the Hopf differential of the Gauss map. We state this result here and provide proof to the next sections.

{cl:existenceuniquew}

Claim 2. Let Φ be a holomorphic quadratic differential on $\mathbb{D}$. Then there exists a unique smooth function $w : \mathbb{D} \to \mathbb{R}$ such that

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1,$$

the metric $e^{2w} ds_p^2$ is complete and $e^{2w} - \|\Phi\|^2 e^{-2w} > 0$.

of Theorem F.1.

1. Well-definedness:
We just observe that by choosing a different $S' \in [S]$ ultimately results to the same map u post-composed with an isometry of the hyperbolic disk $\mathbb{D}$. The Hopf differential associated to u remains unchanged under this transformation.

2. Surjectivity:
Let $[\Phi] \in \mathbf{Q}_{\mathbb{D}}$ and choose a representative

$$\Phi = \varphi(z)\, dz^2.$$

We reconstruct the surface S starting from Φ using the following diagrams

$$\begin{tikzcd}
\Phi \arrow[r, mapsto, "(*1)"] & \|\Phi\| \arrow[r, mapsto, "(*2)"] & w \arrow[r, mapsto, "(*5)"] \arrow[dd, mapsto, "(*3)"] & (h_{ij})_{ij} \arrow[dr, mapsto] & \\
 & & & \text{G.C.} \arrow[r, "(*6)"] & S \\
 & & e^{2w} ds_p^2 \arrow[r, mapsto, "(*4)"] & (g_{ij})_{ij} \arrow[ur, mapsto] &
\end{tikzcd}$$

.

(∗1) Starting from the holomorphic quadratic differential Φ, we compute its norm $\|\Phi\|$ with respect to the Poincaré metric.

(∗2) We recover the function w using Claim 2.

(∗3) We define the bilinear form

$$\mathbf{I} = e^{2\tilde{w}}|dz|^2 = e^{2w} ds_p^2 \ .$$

Which, by the choice of w, it defines a complete metric on $\mathbb{D}$.

(∗4) If the first fundamental form is

$$\mathbf{I} = e^{2\tilde{w}}(dx^2 + dy^2),$$

then its coefficients in the coordinates (x, y) are

$$(g_{ij})_{ij} = e^{2\tilde{w}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} .$$

(∗5) We now define a candidate second fundamental form

$$\mathbf{II} = h_{ij} dx^i dx^j$$

by requiring that

$$\begin{cases} \frac{1}{2}(h_{11} + h_{22}) = e^{2\tilde{w}} \\ h_{11} - h_{22} = 2\Re(\varphi) \\ h_{12} = -\Im(\varphi) \ , \end{cases} \tag{28}$$ {eq:class1}

which are the equations we found in (23). This is a linear system which can be easily solved to find $(h_{ij})_{ij}$.

(∗6) We claim that the pair $(\mathbf{I}, \mathbf{II})$ satisfies the Gauss-Codazzi equations.

For the Codazzi equations, we may recognize that the statement of Theorem D.5 holds not only if Φ is the Hopf differential of a Gauss map, but also by simply assuming that the relation (28) holds.

Regarding Gauss' equation, by Theorem D.6, it is equivalent to w solving the equation

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1 \ ,$$

which we assumed to be true by claim.

Hence $\mathbf{I}$ and$\mathbf{II}$ satisfy the Gauss-Codazzi equations and therefore we can apply Bonnet's Theorem C.1, finding for every point of $\mathbb{D}$ a local immersion in $\mathbb{M}^{2,1}$, whose first and second fundamental forms are $\mathbf{I}$ and $\mathbf{II}$. Since $\mathbb{D}$ is simply connected, we can glue these local immersions to a global one

$$\iota : \mathbb{D} \to \mathbb{M}^{2,1},$$

unique up to post-composition with an isometry of $\mathbb{M}^{2,1}$.
The induced metric of ι is $\mathbf{I} = e^{2w} ds_p^2$, hence complete. Moreover the mean curvature is identically 1, for how we constructed the forms $\mathbf{I}$ and $\mathbf{II}$. Therefore the image surface is a complete spacelike surface in $\mathbb{M}^{2,1}$ with constant mean curvature 1. Then, by Proposition 4, it is an entire graph, hence a hyperbolic CMC-1 cut.

Finally, by construction, the Hopf differential of the Gauss map of the surface we just found is Φ. Therefore

$$\mathscr{A}([S]) = [\Phi],$$

and $\mathscr{A}$ is surjective.

3. Injectivity:
Let $[S_1], [S_2] \in \mathbf{E}$ and assume that

$$\mathscr{A}([S_1]) = \mathscr{A}([S_2]).$$

We want to show that $[S_1] = [S_2]$.
Let us choose some conformal parametrizations

$$\Lambda_i : \mathbb{D} \to S_i, \qquad i = 1, 2,$$

and define

$$\Phi_i = \varphi_i(z)\, dz^2$$

the corresponding quadratic differentials of their Gauss maps. Since we assumed that $\mathscr{A}([S_1]) = \mathscr{A}([S_2])$, up to a change of parameter we may assume that

$$\Phi_1 = \Phi_2 =: \Phi.$$

Then we define w_i be the function associated to S_i as in Theorem E.1. Again from the theorem, we have that

$$\mathbf{I}_i = e^{2w_i} ds_p^2$$

is the first fundamental form of S_i read through Λ_i. Since both surfaces have Hopf differential Φ, the functions w_1 and w_2 solve the same equation

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1,$$

and the metrics

$$e^{2w_1} ds_p^2, \qquad e^{2w_2} ds_p^2$$

are complete. By the Claim 2, it follows that

$$w_1 = w_2.$$

Hence S_1 and S_2 have the same first fundamental form. Moreover, the second fundamental forms are determined by w and Φ through the relations

$$\begin{cases} \frac{1}{2}(h_{11} + h_{22}) & = e^{2\tilde{w}}, \\ h_{11} - h_{22} & = 2\Re(\varphi), \\ h_{12} & = -\Im(\varphi), \end{cases}$$

so that they also coincide.
Therefore the two surfaces have the same first and second fundamental forms on $\mathbb{D}$ and by applying Bonnet's Theorem C.1, the corresponding immersions differ by a post-composition with an isometry of $\mathbb{M}^{2,1}$. In particular, S_1 and S_2 are isometric, hence

$$[S_1] = [S_2]$$

in $\mathbf{E}$. Which proves that $\mathscr{A}$ is injective.

□

G Uniqueness for the PDE

We now begin the proof of Claim 2. Recall that, if Φ is a holomorphic quadratic differential on $\mathbb{D}$ and we set

$$q = \|\Phi\|,$$

the claim asserts the existence and uniqueness of a smooth function $w : \mathbb{D} \to \mathbb{R}$ satisfying

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1, \tag{29}$$ {eq:box}

such that the metric $e^{2w} ds_p^2$ is complete and

$$e^{2w} - q^2 e^{-2w} > 0 \tag{30}$$ {eq:box2}

on $\mathbb{D}$. In this section we prove uniqueness.

G.1 A generalized maximum principle

Let M be a complete Riemannian manifold of dimension n, and fix a point $p \in M$. Assume moreover that M is simply connected and has nonpositive sectional curvature. This assumption is stronger than what is strictly needed in this section, but greatly simplifies the discussion and is satisfied in the case of a complete space-like surface of $\mathbb{M}^{2,1}$. These hypotheses allow us to apply the Cartan-Hadamard theorem, so that the exponential map $\exp_p$ is a diffeomorphism. In particular, this tells us that for every $x \in M$ there exists a unique geodesic joining p to x, and that $d(\cdot,\cdot)$ is smooth on $M \setminus \{x = y\}$.

We need now to define an important quantity we will refer to multiple times later. Let us fix a point $x \in M$ and let σ be the unique unit speed geodesic that joins x to p. In the following expression we also denote with l the length of σ, or in other terms $l = d(x,p)$. Let $\sigma'(t)$ be the velocity vector of σ, and denote with $\mathrm{Ric}(\sigma'(t))$ the Ricci curvature along σ. We then define

$$K(x) = \min_{k\in[0,l]} \left[\frac{n-1}{l-k} - \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \,\mathrm{Ric}(\sigma'(t)) dt \right] \quad . \tag{31}$$ {eq:intromaxprin1}

We now claim that, if $r(x) \geq 1$ and the Ricci curvature is bounded from below, then $K(x)$ is bounded from above by a constant independent of x. Indeed, if we assume that for every unit vector v, $\mathrm{Ric}(v) \geq \mathscr{S}$, then

$$\frac{1}{(l-k)^2} \int_k^l (t-k)^2 \,\mathrm{Ric}(\sigma'(t))\, dt \geq \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \mathscr{S}\, dt = \frac{(l-k)}{3} \mathscr{S} \quad .$$

Therefore in the argument of min in (31) we have

$$\frac{n-1}{l-k} - \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \,\mathrm{Ric}(\sigma'(t))\, dt \leq \frac{n-1}{l-k} - \frac{(l-k)}{3} \mathscr{S} \quad .$$

Since $r(x) = l \geq 1$, if we choose $k = l-1$ we get

$$K(x) \leq n - 1 - \frac{\mathscr{S}}{3} \quad .$$

Which proves our claim.

The next lemma, due to Yau, provides the estimate for the distance function that will be used in the proof of Theorem G.1.

{lm:distest}

Lemma 8. *With the notation introduced above, for every $x \in M \setminus \{p\}$ such that $r(x) \geq 1$ we have*

$$\Delta r(x) \leq K(x) \ .$$

Proof. We redirect the reader to [11] for a proof. □

We are ready to state and prove the maximum principle.

{th:maxprin}

Theorem G.1. *Let M be a complete, simply connected Riemannian manifold of nonpositive sectional curvature, with Ricci curvature bounded from below. Let $f \in C^2(M)$ be bounded from above. Fix $p \in M$, and let $r(x) = d(p, x)$. Then there exist a constant $C > 0$ and a sequence $(x_k)_k \subset M$ such that*

$$f(x_k) \to \sup_M f, \qquad |\nabla f|(x_k) \leq \frac{C}{k}, \qquad \Delta f(x_k) \leq \frac{C}{k}.$$

In particular,

$$\limsup_{k\to\infty} \Delta f(x_k) \leq 0.$$

Proof. Let us define the auxiliary functions $u(x) = (f(x) - f(p) + 1)$ and $L(x) = \log(r^2(x) + 2)$. Then let us fix a positive integer k and define the function g as

$$g(x) = \frac{f(x) - f(p) + 1}{(\log(r^2(x) + 2)^{\frac{1}{k}}} = \frac{u(x)}{L(x)^{\frac{1}{k}}} \quad .$$

Let us observe that $g(p) = \frac{1}{\log(2)^{\frac{1}{k}}} > 0$ and that $\lim_{r(x)\to\infty} g(x) = 0$. Since g is continuous, this means that it attains its maximum at a certain point x_k.

First, let us show that we can assume that $r(x_k) \to \infty$. If that's not the case, up to taking a subsequence, $r(x_k)$ is bounded. Then we can find a closed geodesic ball $\overline{B}(p; \varrho)$ with $\varrho > 0$ that contains x_k for every k sufficiently big. Since M is complete, the Hopf Rinow theorem implies that $\overline{B}(p; \varrho)$ is compact. Therefore, after passing to a subsequence, we may assume that

$$x_k \to x_0 \quad .$$

But we know that $f(x_k) \to \sup f$ and that f is continuous, which implies that x_0 is a maximum point.
This entails that

$$\nabla f(x_0) = 0 \qquad \text{and} \qquad \Delta f(x_0) \leq 0 \quad .$$

Which verifies trivially the thesis by setting $x_k \equiv x_0$. Thus we assume, from now on, that $r(x_k) \to \infty$.

Let us prove first that

$$\limsup_{k\to\infty} f(x_k) = \sup_M f \quad . \tag{32}$$ {eq:boundprin13}

Assume it by contradiction to be false. Then by definition of sup and lim sup there exists a $\delta > 0$ and a $\bar{x} \in M$ such that for k sufficiently big,

$$f(x_k) < f(\bar{x}) - \delta \ , \qquad \text{equivalently,} \qquad f(x_k) + \delta < f(\bar{x}) \quad . \tag{33}$$ {eq:boundprin12}

Since we assumed that $r(x_k) \overset{k\to\infty}{\longrightarrow} \infty$, then we have, for k sufficiently large, $r(x_k) > r(\bar{x})$ and $f(\bar{x}) > f(x_k)$ so that

$$g(\bar{x}) = \frac{f(\bar{x}) - f(p) + 1}{(\log(r^2(\bar{x}) + 2)^{\frac{1}{k}}} > \frac{f(x_k) - f(p) + 1}{(\log(r^2(x_k) + 2)^{\frac{1}{k}}} = g(x_k) \quad .$$

Which contradicts the maximality of x_k. It is obvious that, up to taking a subsequence, we can assume $f(x_k) \to \sup_M f$.

Let us now focus on the two estimates. As we already mentioned, the Cartan-Hadamard M theorem guarantees that the function $d(p,x)^2 = r(x)^2$ is differentiable in x_k.
We can easily calculate the gradient of g,

$$\nabla g(x) = \frac{\nabla f(x)}{\log(r(x)^2 + 2)^{1/k}} - \frac{2(f(x) - f(p) + 1)r(x)}{k(r(x)^2 + 2)\log(r(x)^2 + 2)^{1+1/k}} \nabla r(x) \quad .$$

Since x_k is a maximum point of g_k, we have $\nabla g(x_k) = 0$. Hence

$$\nabla f(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)} \nabla r(x_k) \ , \tag{34}$$ {eq:boundprin9}

$$|\nabla f|(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)} \ , \tag{35}$$ {eq:boundprin8}

where we used $|\nabla r(x_k)| = 1$.
We then observe, since f is bounded from above, also u is

$$u(x) = (f(x) - f(p) + 1) \le \sup_M f - f(p) + 1 =: A.$$

Also, it is not difficult to show that the real function satisfies

$$\frac{2x}{(x^2 + 2)\log(x^2 + 2)} \le 2 \ , \quad . \tag{36}$$ {eq:boundprin14}

so that by substituting in (35) we get the desired estimate

$$|\nabla f|(x_k) \le \frac{2A}{k} \quad .$$

Let us now focus on the estimate on the Laplacian. We can find the expression of the Laplacian of $g = u/L^{\frac{1}{k}}$. First we observe that

$$\begin{aligned}
\Delta g &= \nabla \cdot \nabla \left(\frac{u}{L^{\frac{1}{k}}} \right) \\
&= \nabla \cdot \left(\nabla u L^{-\frac{1}{k}} + u \nabla (L^{-\frac{1}{k}}) \right) \\
&= \frac{\Delta u}{L^{\frac{1}{k}}} + 2 \left\langle \nabla u, \nabla (L^{-\frac{1}{k}}) \right\rangle + u \Delta (L^{-\frac{1}{k}}) \quad ,
\end{aligned} \tag{37}$$ {eq:boundprindeltag1}

then we calculate the involved quantities. First,

$$\nabla u = \nabla f, \qquad \nabla L = \frac{2r\nabla r}{r^2+2}, \qquad \nabla\left(\frac{1}{r^2+2}\right) = -\frac{2r\nabla r}{(r^2+2)^2},$$
$$\nabla\left(L^{-\frac{1}{k}}\right) = -\frac{1}{k}\frac{\nabla L}{L^{\frac{1}{k}+1}} = -\frac{1}{k}\frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+1}},$$
$$\nabla\left(L^{-\frac{1}{k}-1}\right) = \left(-\frac{1}{k}-1\right)\frac{\nabla L}{L^{\frac{1}{k}+2}} = -\frac{k+1}{k}\frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+2}},$$
$$\Delta L = \nabla\cdot\left(\frac{2r\nabla r}{r^2+2}\right) = \frac{2|\nabla r|^2}{r^2+2} + \frac{2r\Delta r}{r^2+2} - \frac{4r^2|\nabla r|^2}{(r^2+2)^2} \quad ,$$

and then

$$\begin{aligned}
\Delta\left(L^{-\frac{1}{k}}\right) &= \nabla\cdot\nabla L^{-\frac{1}{k}} \\
&= -\frac{1}{k}\nabla\cdot\left(\frac{\nabla L}{L^{\frac{1}{k}+1}}\right) \\
&= -\frac{1}{k}\frac{\Delta L}{L^{\frac{1}{k}+1}} - \frac{1}{k}\left\langle \nabla L, \nabla\left(L^{-\frac{1}{k}-1}\right)\right\rangle \\
&= -\frac{1}{k}\left(\frac{2|\nabla r|^2}{r^2+2} + \frac{2r\Delta r}{r^2+2} - \frac{4r^2|\nabla r|^2}{(r^2+2)^2}\right)\frac{1}{L^{\frac{1}{k}+1}} + \frac{k+1}{k^2}\left\langle \frac{2r\nabla r}{r^2+2}, \frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+2}}\right\rangle \\
&= -\frac{2}{k}\frac{1}{(r^2+2)L^{\frac{1}{k}+1}} - \frac{2}{k}\frac{r\Delta r}{(r^2+2)L^{\frac{1}{k}+1}} + \frac{4}{k}\frac{r^2}{(r^2+2)^2L^{\frac{1}{k}+1}} + \frac{4(k+1)}{k^2}\frac{r^2}{(r^2+2)^2L^{\frac{1}{k}+2}} \ .
\end{aligned}$$

Hence, by substituting in (37) and simplifying, we obtain

$$\begin{aligned}
\Delta g = \frac{\Delta f}{L^{1/k}} &- \frac{4r}{k(r^2+2)L^{1+1/k}}\langle\nabla f, \nabla r\rangle - \frac{2u}{k(r^2+2)L^{1+1/k}} \\
&- \frac{2ur\,\Delta r}{k(r^2+2)L^{1+1/k}} + \frac{4ur^2}{k(r^2+2)^2L^{1+1/k}} + \frac{4(k+1)ur^2}{k^2(r^2+2)^2L^{2+1/k}} \ .
\end{aligned}$$

Since x_k is a maximum point of g, we have $\Delta g(x_k) \le 0$. Using the expression for Δg, the identity $\nabla g(x_k) = 0$, and

$$\langle\nabla f, \nabla r\rangle(x_k) = \frac{2u(x_k)r(x_k)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ ,$$

which we have derived from (34), we obtain

$$\begin{aligned}
\Delta f(x_k) \le \frac{2u(x_k)\big(1+r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} &- \frac{4u(x_k)r^2(x_k)}{k(r^2(x_k)+2)^2\log(r^2(x_k)+2)} \\
&- \frac{4(k-1)u(x_k)r^2(x_k)}{k^2(r^2(x_k)+2)^2\log^2(r^2(x_k)+2)} \ .
\end{aligned}$$

In particular, by discarding the last two nonpositive terms, we get

$$\Delta f(x_k) \le \frac{2u(x_k)\big(1+r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ . \tag{38}$$ {eq:boundprin10}

First we find another inequality for a real function, namely

$$\frac{1}{(x^2+2)\log(x^2+2)} \leq 2 \quad . \tag{39}$$ {eq:boundprin20}

, it follows from (38),

$$\begin{aligned}
\Delta f(x_k) &\leq \frac{2u(x_k)\big(1+r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \\
[u \leq A] \qquad &\leq \frac{2A}{k}\left(\frac{1}{(r^2(x_k)+2)\log(r^2(x_k)+2)} + \frac{r(x_k)\Delta r(x_k)}{(r^2(x_k)+2)\log(r^2(x_k)+2)}\right) \\
[(36),\ (39)] \qquad &\leq \frac{2A}{k}\left(2+2\Delta r(x_k)\right) \quad . \qquad (40)
\end{aligned}$$ {eq:boundprin11}

We now apply the estimate of Lemma 8. We can do this because under the assumption that $r(x_k) \to \infty$ we have $r(x_k) \geq 1$ for k sufficienly large, so that

$$\Delta r(x) \leq K(x) \ .$$

But, as we already observed, $K(x)$ is uniformly bounded from above when $r(x) \geq 1$ since the Ricci curvature is bounded from below.
So we conclude in (40) that there is a constant $\tilde{C}$ for which

$$\Delta f(x_k) \leq \frac{\tilde{C}}{k} \ .$$

□

A boundedness result
We will also use this theorem to obtain a boundedness result for a certain differential inequality. We first make a few preliminary observations, and then state the theorem at the end of the section.

Let M be a complete Riemannian manifold with Ricci curvature bounded from below, and let $u \in C^2(M)$ satisfy

$$\Delta u \geq f \quad , \tag{41}$$ {eq:boundprin2}

where the assumptions on f will be specified later.
Now let $F \in C^2(\mathbb{R})$ be a positive function, whose precise form will also be given later, which we require to be strictly increasing or in other terms to have a strictly positive derivative. Since $F > 0$, the function $-1/F$ is bounded from above. We may therefore apply Theorem G.1 to $-1/F(u)$ and obtain a sequence $(x_k)_k$ such that

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = -\sup\left(-\frac{1}{F(u)}\right). \tag{42}$$ {eq:boundprin7}

Moreover, the corresponding gradient and Laplacian estimates hold:

$$\frac{|F'(u(x_k))|\,|\nabla u|(x_k)}{F(u(x_k))^2} \leq \frac{1}{k}, \tag{43}$$ {eq:boundprin3}

$$-\frac{F''(u(x_k))\,|\nabla u|^2(x_k)}{F(u(x_k))^2} - \frac{F'(u(x_k))\,\Delta u(x_k)}{F(u(x_k))^2} + \frac{2(F'(u(x_k)))^2\,|\nabla u|^2(x_k)}{F(u(x_k))^3} \geq -\frac{1}{k}\ . \tag{44}$$ {eq:boundprin16}

For simplicity, from now on we evaluate all quantities at x_k.
Then, multiplying (44) by $\dfrac{F'(u)^2}{F(u)^2|F''(u)|}$, we obtain

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} - \frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} + \frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \geq -\frac{F'(u)^2}{kF(u)^2|F''(u)|}\ . \tag{45}$$ {eq:boundprin1}

We now estimate the terms on the left-hand side of (45). First,

$$-\frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} \leq -\frac{F'(u)^3}{F(u)^4}\frac{f(u)}{|F''(u)|}\ . \qquad \text{[by (41), } F'>0\text{]}$$

Moreover,

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \leq \left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \leq \frac{1}{k^2}\ . \qquad \text{[by (43)]}$$

Also,

$$\begin{aligned}\frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} &\leq 2\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \\ &\leq \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\ . \qquad \text{[by (43)]}\end{aligned}$$

Therefore, from (45) we obtain

$$\frac{1}{k^2} - \frac{F'(u)^3 f(u)}{F(u)^4|F''(u)|} + \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right| \geq -\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\ . \tag{46}$$ {eq:boundprin6}

Suppose now that F satisfies

$$\limsup_{t\to\infty}\frac{|F'(t)|^2}{F(t)\,|F''(t)|} < \infty\ , \tag{47}$$ {eq:boundprin4}

and

$$\liminf_{t\to\infty}\frac{F'(t)^3 f(t)}{F(t)^4\,|F''(t)|} > 0\ . \tag{48}$$ {eq:boundprin5}

We claim that this implies

$$\limsup_{k\to\infty} u(x_k) < \infty\ .$$

Indeed, suppose by contradiction that $\limsup_{k\to\infty} u(x_k) = \infty$. Passing to a subsequence if necessary we may assume that

$$u(x_k) \to \infty\ .$$

Taking $\limsup$ in (46), and using (47) together with the fact that later our choice of F will satisfy $F \geq 1$, we obtain

$$-\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \geq 0 \quad ,$$

that is,

$$\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \leq 0 \quad ,$$

which contradicts (48).

So we have proved so far that

$$\limsup_{k\to\infty} u(x_k) < \infty \quad .$$

Passing to a subsequence if necessary, we may therefore assume that

$$u(x_k) \to \ell$$

for some $\ell \in \mathbb{R}$.

Let us now show that u is bounded from above.
Since $F > 0$, from (42) we obtain

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = \frac{1}{\sup_M F(u)} \quad ,$$

and therefore

$$F(u(x_k)) \to \sup F(u) \quad .$$

On the other hand, by continuity of F and the fact that $u(x_k) \to \ell$, we also have

$$F(u(x_k)) \to F(\ell) \quad .$$

Hence

$$F(\ell) = \sup F(u) \quad .$$

Since F is strictly increasing, for every $x \in M$ we have

$$F(u(x)) \leq F(\ell) \qquad \Longrightarrow \qquad u(x) \leq \ell \quad .$$

This means that ℓ is an upper bound for u, hence u is bounded from above:

$$\sup u < \infty \quad .$$

We have therefore reduced the problem to finding a function F which is positive, strictly increasing, and satisfies (47) and (48). Let us search for such a function of the form

$$F(s) = \int_a^s \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt + 1 \quad , \tag{49}$$ {eq:boundprin15}

where g is a positive continuous function defined on $[a, \infty)$ and such that

$$\int_b^\infty \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt < \infty$$

for some $b \geq a$. If we want F to be positive, the expression in (49) may fail to achieve that on the whole line. So we define it on $[a, \infty)$ and then extend F to all of $\mathbb{R}$ in such a way that it becomes a strictly increasing C^2 function.

Let us now translate conditions (47) and (48) into conditions on the function g. To do so, we first compute the relevant quantities.

$$F'(s) = \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad ,$$

and therefore

$$F''(s) = -\frac{1}{2} \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{3}{2}} g(s) \quad .$$

Hence

$$\frac{|F'(t)|^2}{F(t)\,|F''(t)|} = 2\, \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} \quad ,$$

so condition (47) is equivalent to

$$\limsup_{t \to \infty} \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} < \infty \quad .$$

Moreover,

$$\frac{F'(t)^3 f(t)}{F(t)^4 |F''(t)|} = 2\, \frac{f(t)}{F(t)^4 g(t)} \quad ,$$

so condition (48) is equivalent to

$$\liminf_{t \to \infty} \frac{f(t)}{F(t)^4 g(t)} > 0 \quad .$$

Finally, we observe that since g is positive, if $s \leq t$, then

$$\left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \geq \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad .$$

Therefore, for $t \geq b$,

$$\begin{aligned} F(t) - 1 &= \int_a^t \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &\geq \int_b^t \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &\geq \int_b^t \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &= (t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad . \end{aligned}$$

Hence

$$F(t) \geq (t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad .$$

This allows us to further estimate the quantity appearing in (47):

$$\begin{aligned} \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} &\leq \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{(t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-1/2} g(t)} \\ &= \frac{\displaystyle\int_a^t g(\tau)\, d\tau}{(t-b) g(t)} \\ &\sim \frac{\displaystyle\int_a^t g(\tau)\, d\tau}{t\, g(t)} \quad . \end{aligned}$$

It is convenient to rewrite what we found in terms of the primitive of g.Let

$$G(t) = \int_a^t g(\tau)\, d\tau,$$

then $G \in C^1([a, \infty))$, $G(a) = 0$, and $G'(t) = g(t) > 0$ for every $t \geq a$. In these terms, the previous discussion yields the following theorem.

{th:boundprin}

Theorem G.2. *Let M be a Riemannian manifold whose Ricci curvature is bounded from below, and let $u \in C^2(M)$ satisfy*

$$\Delta u \geq f(u) \quad .$$

A sufficient condition for u to be bounded from above on M is the existence of $a \in \mathbb{R}$ and of an increasing positive function $G \in C^1([a, \infty))$ such that

$$\begin{cases} \displaystyle\int_b^\infty G(t)^{-1/2}\, dt < \infty \text{ for some } b \geq a \\ \displaystyle\limsup_{t \to \infty} \frac{G(t)}{t\, G'(t)} < \infty \\ \displaystyle\liminf_{t \to \infty} \frac{f(t)}{G'(t)} > 0 \ . \end{cases}$$

G.2 The uniqueness theorem

We now prove the uniqueness part of Claim 2. We recall that, if

$$w : \mathbb{D} \to \mathbb{R}$$

solves the hyperbolic equation

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1,$$

then the function

$$\tilde{w} = w + \log\left(\frac{2}{1-|z|^2}\right) \tag{50}$$ {eq:unithm6}

solves an equivalent equation with respect to the Euclidean background metric. Therefore it is enough to prove uniqueness in the Euclidean formulation and then translate the conclusion back to the original one.

Theorem G.3. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$. Let $\tilde{w}$ be a solution of*

$$\Delta\tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \tag{51}$$ {eq:unithm}

in Ω, and assume that

$$\Delta\tilde{w} \geq 0 \quad ,$$

and that $e^{2\tilde{w}}|dz|^2$ is a complete metric on Ω. Then any other solution v of the equation satisfies $v \leq \tilde{w}$.

Proof. Let us define $\eta = v - \tilde{w}$. We need to prove that $\eta \leq 0$.
We have

$$\begin{aligned}\Delta\eta &= \Delta v - \Delta\tilde{w} \\ &\overset{[(51)]}{=} e^{2v} - h^2 e^{-2v} - e^{2\tilde{w}} + h^2 e^{-2\tilde{w}} \quad .\end{aligned} \tag{52}$$ {eq:unithm2}

Let us now try to explicit $\tilde{w}$ and η in the RHS of (52),

$$e^{2v} = e^{2(\eta+\tilde{w})} = e^{2\eta} e^{2\tilde{w}} \quad .$$

Let us define the quantity $|\mu|^2 = h^2 e^{-4\tilde{w}}$, we have then

$$h^2 e^{-2v} = h^2 e^{-2(\eta+\tilde{w})} = h^2 e^{-2\tilde{w}} e^{-2\eta} = |\mu|^2 e^{2\tilde{w}} e^{-2\eta},$$

and

$$h^2 e^{-2\tilde{w}} = \left(h^2 e^{-4\tilde{w}}\right) e^{2\tilde{w}} = |\mu|^2 e^{2\tilde{w}} \quad .$$

So, by substituting what we just found in (52), we get

$$\begin{aligned}\Delta\eta &= e^{2\tilde{w}} e^{2\eta} - |\mu|^2 e^{2\tilde{w}} e^{-2\eta} - e^{2\tilde{w}} + |\mu|^2 e^{2\tilde{w}} \\ &= e^{2\tilde{w}}(e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2) \quad .\end{aligned} \tag{53}$$ {eq:unithm3}

The Laplacian with respect to the metric $g = e^{2\tilde{w}}|dz|^2$ is simply $e^{-2\tilde{w}}\Delta$. So in (53) we obtain

$$\Delta_g \eta = e^{-2\tilde{w}}\Delta\eta = e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \quad . \tag{54}$$ {eq:unithm5}

Let us now observe that $|\mu|^2 \leq 1$. Indeed, by (51),

$$e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \quad ,$$

hence

$$e^{2\tilde{w}}(1 - h^2 e^{-4\tilde{w}}) \geq 0 \quad ,$$

that is,

$$1 - |\mu|^2 \geq 0 \quad ,$$

so $|\mu|^2 \leq 1$.
Therefore, from (54) we get

$$\begin{aligned} \Delta_g \eta &= e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \\ &\geq e^{2\eta} - e^{-2\eta} - 1 =: f(\eta) \quad . \end{aligned}$$

We now apply Theorem G.2 to the function η on the complete simply connected Riemannian surface (Ω, g). For the auxiliary function appearing in the theorem, we may take

$$\gamma(t) = t^2$$

on $[1, \infty)$. A direct computation shows that γ satisfies the required assumptions, since

$$\int_b^\infty \left(\int_1^t \tau^2 \, d\tau \right)^{-1/2} dt < \infty, \qquad \limsup_{t\to\infty} \frac{\int_1^t \tau^2 \, d\tau}{t^3} < \infty,$$

and

$$\liminf_{t\to\infty} \frac{f(t)}{t^2} = \liminf_{t\to\infty} \frac{e^{2t} - e^{-2t} - 1}{t^2} > 0 \quad .$$

It remains only to check that the Ricci curvature of g is bounded from below which, since we are on a surface, is the same as the Gaussian curvature. For the metric

$$g = e^{2\tilde{w}}|dz|^2 \quad ,$$

the Gaussian curvature is

$$\begin{aligned} K(g) &= -e^{-2\tilde{w}}\Delta\tilde{w} \\ &= -e^{-2\tilde{w}}\left(e^{2\tilde{w}} - h^2 e^{-2\tilde{w}}\right) \\ &= -1 + h^2 e^{-4\tilde{w}} \\ &= -1 + |\mu|^2 \geq -1 \quad . \end{aligned} \tag{55}$$ {eq:unithm10}

Since $|\mu|^2 \leq 1$, it follows that

$$-1 \leq K(g) \leq 0 \quad ,$$

and in particular, looking at (55), the curvature is bounded from below. Therefore we can apply Theorem G.2 and conclude that η is bounded from above. Now we can also apply Theorem G.1 to η, obtaining a sequence $(z_k)_k \subset \Omega$ such that

$$\eta(z_k) \to \bar{\eta} = \sup_{\Omega} \eta, \qquad \limsup_{k\to\infty} \Delta_g \eta(z_k) \leq 0 \quad .$$

Since $|\mu|^2$ is continuous and bounded, by passing to a subsequence if necessary we may also assume that

$$|\mu|^2(z_k) \to a$$

for some $a \leq 1$. Taking limits in (54), we obtain

$$\begin{aligned} 0 &\geq \limsup_{k\to\infty} \Delta_g \eta(z_k) \\ &= \lim_{k\to\infty} \left(e^{2\eta(z_k)} - |\mu|^2(z_k) e^{-2\eta(z_k)} - 1 + |\mu|^2(z_k) \right) \\ &= e^{2\bar{\eta}} - a e^{-2\bar{\eta}} - 1 + a \\ &=: \varphi(\bar{\eta}) \quad . \end{aligned} \tag{56}$$

{eq:unithm4}

We have defined in the last expression the function $\varphi(s) = e^{2s} - ae^{-2s} - 1 + a$, which has strictly positive derivative $\varphi'(s) = 2e^{2s} + 2ae^{-2s} > 0$, so that it is strictly increasing. Moreover $\varphi(0) = 0$, hence (56) implies

$$\bar{\eta} \leq 0, \qquad \text{therefore, on } \Omega, \qquad \eta \leq 0 \quad .$$

This finally implies $v \leq \tilde{w}$. □

{cor:uni1}

Corollary G.1. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$ and let $\tilde{w}_1, \tilde{w}_2$ two solutions of*

$$\Delta \tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \tag{57}$$

in Ω such that $e^{2\tilde{w}_i}$ are complete metrics on Ω. Then $\tilde{w}_1 \equiv \tilde{w}_2$.

Proof. Just observe that for the previous theorem, $\tilde{w}_1 \leq \tilde{w}_2$ and $\tilde{w}_2 \leq \tilde{w}_1$. □

{cor:uni2}

Corollary G.2. *There is at most one solution of the equation* (29) *such that the associated metric is complete.*

Proof. Let w_1, w_2 be two solutions of

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1$$

satisfying (30), and assume that the metrics $e^{2w_1} ds_p^2$ and $e^{2w_2} ds_p^2$ are complete.

For $i = 1, 2$, define

$$\tilde{w}_i = w_i + \log\left(\frac{2}{1 - |z|^2} \right) \quad .$$

Then

$$e^{2\tilde{w}_i}|dz|^2 = e^{2w_i} ds_p^2,$$

hence the metric $e^{2\tilde{w}_i}|dz|^2$ is complete, as it is the same metric as $e^{2w_i} ds_p^2$, assumed to be complete by hypothesis.

Now, set

$$h(z) = \frac{4q(z)}{(1-|z|^2)^2}.$$

By recalling the discussion in Section D.6 the equation can be rewritten with the Euclidean background metric so that $\tilde{w}_i$ solves

$$\Delta \tilde{w}_i = e^{2\tilde{w}_i} - h^2 e^{-2\tilde{w}_i}.$$

Moreover, condition (30) is equivalent to

$$e^{2\tilde{w}_i} - h^2 e^{-2\tilde{w}_i} > 0,$$

and so

$$\Delta \tilde{w}_i \geq 0.$$

Therefore $\tilde{w}_1$ and $\tilde{w}_2$ satisfy the assumptions of Corollary G.1, so that

$$\tilde{w}_1 = \tilde{w}_2.$$

Which finally implies that

$$w_1 = w_2 \quad .$$

□

H Existence for the PDE

We now prove the existence part of Claim 2. After exposing briefly some preliminaries, the proof proceeds in three steps: first we recall a compact-domain existence theorem based on sub- and super-solutions, then we generalize it to the case of noncompact domains, and finally we give yet another generalization for non bounded data.

H.1 A compact-domain existence theorem

{th:subsupsol_bddom}

Theorem H.1 (Sub-/super-solution method on bounded domains.)**.** *Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with smooth boundary, and let $F \in C^\infty(\overline{\Omega} \times \mathbb{R})$ satisfy*

$$F_u(x,s) > 0 \qquad \text{for all } (x,s) \in \overline{\Omega} \times \mathbb{R}.$$

Assume that there exist

$$\underline{\psi}, \overline{\psi} \in C^0(\overline{\Omega}) \cap H^1(\Omega), \qquad \underline{\psi} \leq \overline{\psi} \text{ on } \overline{\Omega},$$

such that on Ω,

$$\begin{cases} \Delta \overline{\psi} \leq F(x, \overline{\psi}) \\ \Delta \underline{\psi} \geq F(x, \underline{\psi}) \end{cases} \qquad \text{weakly.}$$

Let $\varphi \in C^0(\partial\Omega)$ satisfy

$$\underline{\psi} \le \varphi \le \overline{\psi} \qquad \text{on } \partial\Omega.$$

Then there exists

$$u \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$$

such that

$$\Delta u = F(x,u) \quad \text{in } \Omega, \qquad u = \varphi \quad \text{on } \partial\Omega,$$

and

$$\underline{\psi} \le u \le \overline{\psi} \qquad \text{on } \overline{\Omega}.$$

An important observation is that, for our purposes, the bounded-domain theorem can be applied without substantial changes on a Riemann surface endowed with a conformal metric. More precisely, let

$$g = e^{2\eta}|dz|^2$$

on a relatively compact smooth domain $\Omega \Subset \mathbb{D}$. Then the equation

$$\Delta u = F(x,u)$$

is equivalent to

$$\Delta_g u = \widetilde{F}(x,u), \qquad \widetilde{F}(x,s) := e^{-2\eta}F(x,s).$$

Indeed, in conformal coordinates one has

$$\Delta_g = e^{-2\eta}\Delta.$$

Moreover, the structural assumptions are preserved:

- Differentiation with respect to the unknown variable is unaffected by the conformal factor, hence

$$\partial_s \widetilde{F}(x,s) = e^{-2\eta}\partial_s F(x,s) > 0.$$

- A subsolution or supersolution for the Euclidean equation is also a subsolution or supersolution for the equation written with respect to g, and conversely. For example,

$$\begin{aligned}\Delta\overline{\psi} \le F(x,\overline{\psi}) \text{ weakly} &\Leftrightarrow e^{-2\eta}\Delta\overline{\psi} \le e^{-2\eta}F(x,\overline{\psi}) \text{ weakly} \\ &\Leftrightarrow \Delta_g\overline{\psi} \le \widetilde{F}(x,\overline{\psi}) \text{ weakly }.\end{aligned}$$

Therefore the bounded-domain sub-supersolution theorem applies in exactly the same way on (Ω, g).

Theorem H.2. *Let M be a complete Riemannian manifold, and $F(x,u)$ be a C^∞-function on $M \times \mathbb{R}$ such that*

$$F_u > 0.$$

If there exists $\psi_- < \psi \in C^0(M) \cap H^1(M)$ bounded such that

$$\begin{aligned}\Delta\psi &< F(x,\psi) \\ \Delta\psi_- &> F(x,\psi_-)\end{aligned}$$

weakly, then there exists $u \in C^\infty(M)$ such that

$$\Delta u = F(x, u)$$

and

$$\psi_- < u < \psi,$$

where ψ_- and ψ are called the sub- and super-solutions respectively.

Proof. Fix $k_0 \in \mathbb{N}$ and set

$$v_k := u_{k_0+k}$$

for $k \geq 1$. Then v_k is defined on D_{k_0+k}, hence in particular on D_{k_0+1}, and satisfies

$$\Delta v_k = F(x, v_k)$$

on D_{k_0+1}. Moreover, since each u_j is trapped between the same subsolution and supersolution, we have

$$\psi_- \leq v_k \leq \psi$$

on D_{k_0+1} for every $k \geq 1$.

Since D_{k_0+1} is relatively compact and ψ_-, ψ are continuous, there exists a constant $M > 0$ such that

$$|v_k| \leq M$$

on D_{k_0+1} for every k. Since F is smooth, it is bounded on the compact set $\overline{D_{k_0+1}} \times [-M, M]$, hence there exists $C_0 > 0$ such that

$$|F(x, v_k(x))| \leq C_0$$

for every $x \in D_{k_0+1}$ and every k. Equivalently,

$$|\Delta v_k| \leq C_0$$

on D_{k_0+1}.

Now choose a smooth domain Ω_1 such that

$$D_{k_0} \Subset \Omega_1 \Subset D_{k_0+1}.$$

Fix $p > 2$. By the interior $W^{2,p}$ estimate for the Laplacian, there exists a constant $C_1 > 0$, independent of k, such that

$$\|v_k\|_{W^{2,p}(\Omega_1)} \leq C_1 \Big(\|v_k\|_{L^p(D_{k_0+1})} + \|\Delta v_k\|_{L^p(D_{k_0+1})} \Big).$$

The right-hand side is uniformly bounded in k, because

$$\|v_k\|_{L^p(D_{k_0+1})} \leq M \, |D_{k_0+1}|^{1/p}$$

and

$$\|\Delta v_k\|_{L^p(D_{k_0+1})} \leq C_0 \, |D_{k_0+1}|^{1/p}.$$

Therefore there exists $C_2 > 0$ such that

$$\|v_k\|_{W^{2,p}(\Omega_1)} \leq C_2$$

for every k.

Since $\dim M = 2$, the Sobolev embedding gives

$$W^{2,p}(\Omega_1) \hookrightarrow C^{1,\beta}(\overline{\Omega_1}), \qquad \beta = 1 - \frac{2}{p}.$$

Choosing p sufficiently large, we may assume that $\beta > \alpha$. Hence there exists $C_3 > 0$ such that

$$\|v_k\|_{C^{1,\beta}(\Omega_1)} \leq C_3$$

for every k.

We next show that $F(\cdot, v_k)$ is uniformly bounded in $C^{0,\beta}(\Omega_1)$. Since F is smooth, its derivatives with respect to both variables are bounded on the compact set $\overline{\Omega_1} \times [-M, M]$. Thus there exist constants $A, B > 0$ such that for every $x, y \in \Omega_1$,

$$\begin{aligned} |F(x, v_k(x)) - F(y, v_k(y))| &\leq |F(x, v_k(x)) - F(y, v_k(x))| + |F(y, v_k(x)) - F(y, v_k(y))| \\ &\leq A|x - y| + B|v_k(x) - v_k(y)|. \end{aligned}$$

Using the uniform $C^{0,\beta}$ bound on v_k, we obtain

$$|F(x, v_k(x)) - F(y, v_k(y))| \leq A|x - y| + BC_3|x - y|^\beta.$$

Since Ω_1 is bounded and $\beta < 1$, the first term is also controlled by a constant multiple of $|x - y|^\beta$. Hence there exists $C_4 > 0$ such that

$$\|F(\cdot, v_k)\|_{C^{0,\beta}(\Omega_1)} \leq C_4$$

for every k.

Now choose another smooth domain Ω_0 such that

$$D_{k_0} \Subset \Omega_0 \Subset \Omega_1.$$

Applying the interior Schauder estimate to

$$\Delta v_k = F(\cdot, v_k)$$

on Ω_1, we get a constant $C_5 > 0$, independent of k, such that

$$\|v_k\|_{C^{2,\beta}(\Omega_0)} \leq C_5 \Big(\|v_k\|_{C^0(\Omega_1)} + \|F(\cdot, v_k)\|_{C^{0,\beta}(\Omega_1)} \Big).$$

The right-hand side is uniformly bounded in k by the estimates above, so there exists $C_6 > 0$ such that

$$\|v_k\|_{C^{2,\beta}(\Omega_0)} \leq C_6$$

for every k. In particular,

$$\|v_k\|_{C^{2,\beta}(D_{k_0})} \leq C_6.$$

Since $\beta > \alpha$, the inclusion $C^{2,\beta}(D_{k_0}) \hookrightarrow C^{2,\alpha}(D_{k_0})$ is continuous. Therefore there exists $C_7 > 0$, independent of k, such that

$$\|v_k\|_{C^{2,\alpha}(D_{k_0})} \leq C_7$$

for every k.

Recalling that $v_k = u_{k_0+k}$, this proves that the sequence $(u_{k_0+k})_k$ is bounded in $C^{2,\alpha}(D_{k_0})$. □

Proof. todo As we have already discussed, we may suppose without loss of generality that $M = \mathbb{R}^2$.
Let us fix an increasing relatively compact sequence of open bounded domains $(D_k)_{k=1}^{\infty}$ with sufficiently regular boundary that invade M, that is

$$\overline{D_k} \subset D_{k+1}, \qquad \bigcup_{k=1}^{\infty} D_k = \mathbb{R}^2 \ .$$

We may choose, for every m a boundary data ψ_m on ∂D_m in between $\underline{\psi}$ and $\overline{\psi}$, in other words

$$\underline{\psi} \leq \psi_m \leq \overline{\psi} \ .$$

Then we apply theorem H.1 to the data $D_m, \psi_m, \underline{\psi}, \overline{\psi}$ to find a smooth solution u_m that verifies

$$\underline{\psi} \leq u_m \leq \overline{\psi} \quad \text{on } \overline{D_m}.$$

Then we fix $k_0, k \in \mathbb{N}$, $k > 0$ and set $m = k_0 + k$. Since u_{k_0+k} satisfies $\Delta u_{k_0+k}(x) = F(x, u_{k_0+k})$ on $D_{k_0+k} \supset D_{k_0}$, we may apply the Schauder interior estimate on $D_{k_0} \subset D_k$ to obtain the inequality

$$\|u_{k_0+k}\|_{C^{1,\alpha}(D_{k_0})} \leq C \left(\|F(\,\cdot\,, u_{k_0+k})\|_{C^0(D_{k_0+k})} + \|u_{k_0+k}\|_{L^2(D_{k_0+k})} \right) \ . \tag{58}$$ {eq:thmnoncptex1}

Now we observe,

- ▷ For every $x \in D_{k_0+k}$,

$$|u_{k_0+k}(x)| \leq \max\{|\underline{\psi}|, |\overline{\psi}|\}(x) \leq C$$

  (because $\underline{\psi}, \overline{\psi}$ are bounded),
- ▷ then, since $F_u > 0$ and so F is strictly increasing, we have

$$F(x, u_{k_0+k}(x)) \leq F(x, \max\{|\underline{\psi}|, |\overline{\psi}|\}(x)) \leq C \ ,$$

  (again because of continuity).

Then, by substituting the inequalities we just found in (58), we obtain

$$\|u_{k_0+k}\|_{C^{2,\alpha}(D_{k_0})} \leq C \ .$$

Fix now a compact set $K \subset M$. For k_0 sufficiently large and for every $k > 0$ there exists a D_{k_0} that contains K and such that

$$\|u_{k_0+k}\|_{C^2(K)} \leq \|u_{k_0+k}\|_{C^2(D_{k_0})} \leq C \ .$$

Hence we can apply the Ascoli-Arzelà Theorem for $C^{\alpha,s}$ spaces, obtaining a subsequence $(u_{k'})_{k'}$ that converges in $C^2(K)$.

This construction works for every compact set K, so using a diagonal argument we find a limiting function defined on all of M which is C^2 and solves the equation $\Delta u = F(x,u)$.

Since $F_u > 0$ we can apply the boostrapping argument and obtain that $u \in C^\infty$.tbd □

H.2 The bounded case

We may prove the existence theorem with the stronger hypothesis of boundedness.

{pr:exboundcase}

Proposition 6. *Let Φ a holomorphic quadratic differential on $\mathbb{D}$, such that $\|\Phi\|$ is bounded. Then the equation* **??** *has a unique C^∞ solution w that satisfies*

$$1 \leq e^{2w} \leq \frac{1+\sqrt{1+4a^2}}{2} \ ,$$

with $a = \sup \|\Phi\|$.

Proof. We use the method of sub- and super-solutions, applied to the equation in the form with respect to the metric, for simplicity.

□

H.3 The general case

Theorem H.3. *Let Φ be a holomorphic quadratic differential on $\mathbb{D}$, then* **??** *has a unique C^∞ solution.*

Proof. The idea of this proof is to construct a sequence of solutions on consecutive domains that exhaust $\mathbb{D}$ and prove their convergence in a topology sufficient to obtain the equation valid for the limit.

For every positive integer k, denoting with $d_p(\cdot,\cdot)$ the Poincaré metric on $\mathbb{D}$, define the sequence of open bounded sets

$$\begin{aligned} D_k &:= \{z \in \mathbb{D} \mid d_p(z,0) \leq k\} \\ &= \{z \in \mathbb{D} \mid |z| \leq R_k\} \ , \end{aligned}$$

where $R_k = \frac{e^k-1}{e^k+1} < 1$. Then

$$\overline{D_k} \subset D_{k+1} \text{ for every } k, \quad \bigcup_{k=1}^{\infty} D_k = \mathbb{D} \ .$$

Every D_k can be seen as a Riemannian manifold, with the metric

$$ds_{p,k} = \frac{4R_k^2}{(R_k - |z|)^2}|dz|^2 = \lambda_{p,k}^2 |dz|^2 \ .$$

Furthermore $\overline{D_k}$ is compact and $|\Phi|$ is smooth, then $|\Phi|$ is bounded.
Hence we may apply 6 and find, for every k a function $\tilde{w}_k$ that solves

$$\begin{cases} \Delta \tilde{w}_k = e^{2\tilde{w}_k} - |\Phi|^2 e^{-2\tilde{w}:k} \geq 0 \\ (D_k, e^{2\tilde{w}_k}|dz|^2) \text{ is complete.} \end{cases}$$

It may be useful to define w_k the conformal factor with respect to the Euclidean metric $|dz|^2$. So that

$$\begin{aligned} \tilde{w}_k &= w_k + \log(\lambda_{p,k}) \\ &= w_k + \log\left(\frac{2R_k}{R_k^2 - |z|^2}\right) . \end{aligned}$$

In the proof of Proposition 6 we have seen that $e^{2w_k} \geq 1$, this translates on $\tilde{w}_k$ to

$$e^{2\tilde{w}_k} = e^{2w_k} \frac{4R_k^2}{(R_k^2 - |z|^2)^2} \geq \frac{4R_k^2}{(R_k^2 - |z|^2)^2} \geq \frac{4}{(1 - |z|^2)^2} ,$$

where in the last inequality we used that $\frac{R_k^2}{(R_k^2 - r^2)} \geq \frac{1}{1-r^2}$, which can be proved using $0 < r < R_k < 1$.

Now, again

□

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